NSW Education Standards Authority

Mathematics K–10 (2022)

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Mathematics K–10 (2022)

Overview

Syllabus overview

Organisation of Mathematics K–10

The syllabus structure illustrates the important role Working mathematically plays across all areas of mathematics and reflects the strengthened connections between concepts. Working mathematically has been embedded in the outcomes, content and examples of the syllabus.

Mathematics K–10 outcomes and their related content are organised in:

- Number and algebra
- Measurement and space
- Statistics and probability

Working mathematically

The Working mathematically processes present in the Mathematics K–10 syllabus are:

- communicating
- understanding and fluency
- reasoning
- problem solving.

Students learn to work mathematically by using these processes in an interconnected way. The coordinated development of these processes results in students becoming mathematically proficient.

When students are Working mathematically it is important to help them to reflect on how they have used their thinking to solve problems. This assists students to develop '*mathematical habits of mind' (Cuoco et al. 2010).*

Students need many experiences that require them to relate their knowledge to the vocabulary and conceptual frameworks of mathematics.

Overarching Working mathematically outcome

To highlight how these processes are interrelated, in Mathematics K–10 there is one overarching Working mathematically outcome.

A student develops understanding and fluency in mathematics through:

- exploring and connecting mathematical concepts
- choosing and applying mathematical techniques to solve problems
- communicating their thinking and reasoning coherently and clearly.

The Working mathematically outcome describes the thinking and doing of mathematics. In doing so, the outcome indicates the breadth of mathematical actions that teachers need to emphasise. The overarching Working mathematically outcome is the same across the K–10 Mathematics syllabus.

The Working mathematically processes should be embedded within the concepts being taught. Embedding Working mathematically ensures students are able to fluently understand concepts and

make connections to other focus areas. The mathematics focus area outcomes and content provide the knowledge and skills for students to reason about, and contexts for problem solving. The overarching Working mathematically outcome is assessed in conjunction with the mathematics content outcomes. The sophistication of Working mathematically processes develops through each stage of learning and can be observed in relation to the increase in complexity of the mathematics outcomes and content. A student's level of competence in Working mathematically can be monitored over time, for example, within Additive Relations by the choice of strategy appropriate to the task, and the use of efficient strategy for the stage of learning the student is working at.

Further information is available in Elaborating on Working mathematically in K–10 (Word, 5 pages, 914.28 KB).

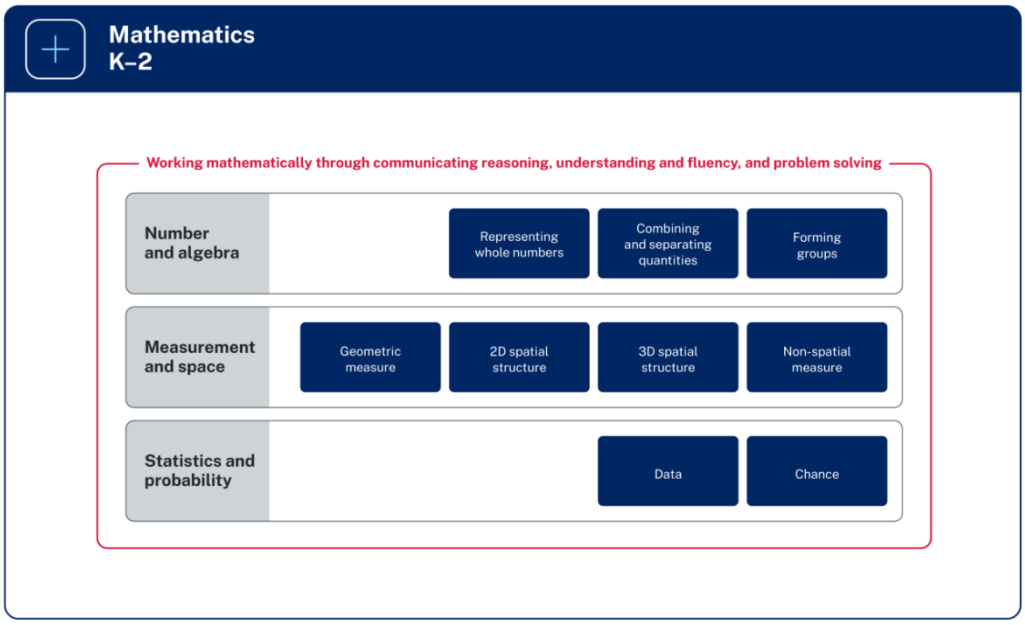


Figure 1: The organisation of Mathematics K–2

Image long description: An overview of the syllabus structure for Early Stage 1 and Stage 1 in Mathematics across the 3 areas of Number and algebra, Measurement and space, and Statistics and probability. Number and algebra reads horizontally across Representing whole numbers, Combining and separating quantities, and Forming groups. Measurement and space reads horizontally across Geometric measure, 2D spatial structure, 3D spatial structure, and Non-spatial measure. Statistics and probability reads horizontally across Data and Chance.

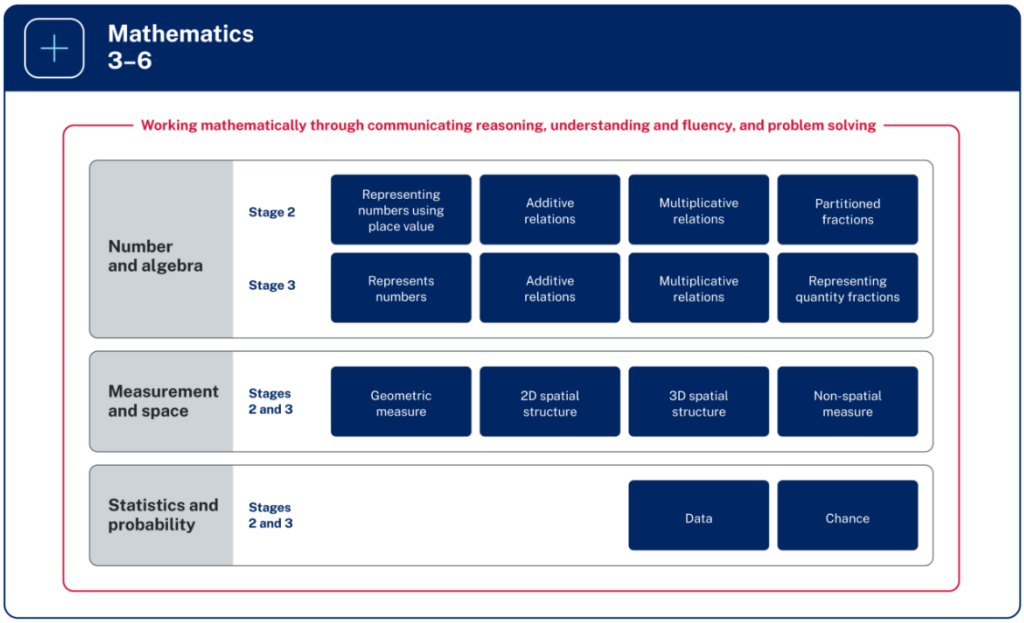


Figure 2: The organisation of Mathematics 3–6

Image long description: An overview of the syllabus structure for Stages 2 and 3 in Mathematics across the 3 areas of Number and algebra, Measurement and space, and Statistics and probability. Number and algebra reads horizontally across 2 stages – Stage 2 and Stage 3. Stage 2 learning areas include Representing numbers using place value, Additive relations, Multiplicative relations and Partitioned fractions. Stage 3 learning areas include Represents numbers, Additive relations, Multiplicative relations, and Representing quality fractions. Measurement and space reads horizontally across 2 stages – Stages 2 and 3. Learning areas include Geometric measure, 2D spatial structure, 3D spatial structure, and Non-spatial measure. Statistics and probability reads horizontally across 2 stages – Stages 2 and 3. Learning areas include Data and Chance.

K–6 Parts A and B

Across a stage of learning all focus areas need to be taught. Mathematics focus areas outline the development of several concepts. In Mathematics K–6, where stages span 2 years of learning (for example, Stage 2 includes Year 3 and Year 4), there are concepts that may need to be addressed earlier or later in the stage.

To assist programming, the content in these focus areas has been separated into 2 parts, A and B, such as in Representing Numbers Using Place Value – A and Representing Numbers Using Place Value – B:

- Part A typically focuses on early concept development
- Part B builds on these early concepts.

The content across Parts A and B relates to the same stage-based outcomes. Teachers can choose which content from Part A and/or Part B to address, based on students' prior learning, needs and abilities.

For example, in Stage 2, Part A does not equate to Year 3 only. When teaching a Year 4 class, the teacher may need to address or consolidate some concepts within Part A prior to addressing concepts in Part B. Similarly, when teaching a Year 3 class, the teacher may decide to address concepts in Part B based on the students' prior learning, needs and abilities.

The Part A and Part B structure of the content:

- provides flexibility for teachers in planning teaching and learning programs based on the needs and abilities of students
- helps to better visualise the progression and growth of concepts within a stage of learning
- makes clear how content builds to support deep understanding in each focus area.

Considerations for planning teaching and learning programs include:

- when students may have learnt some concepts from Part B content in the first year of a stage, consolidation of these concepts in the second year of a stage may be needed
- revisiting concepts regularly to build deeper understanding of mathematical concepts
- providing extension of certain concepts based on students' needs and abilities.

Access content points K–6

Access content points have been developed to support students with significant intellectual disability who are working towards Early Stage 1 outcomes. These students may communicate using verbal and/or nonverbal forms.

For each of the Early Stage 1 outcomes, access content points are provided to indicate content that students with significant intellectual disability may access as they work towards the outcomes. Teachers will use the access content points on their own, or in combination with the content for each outcome.

Decisions regarding curriculum options for students with disability should be made in the context of collaborative curriculum planning.

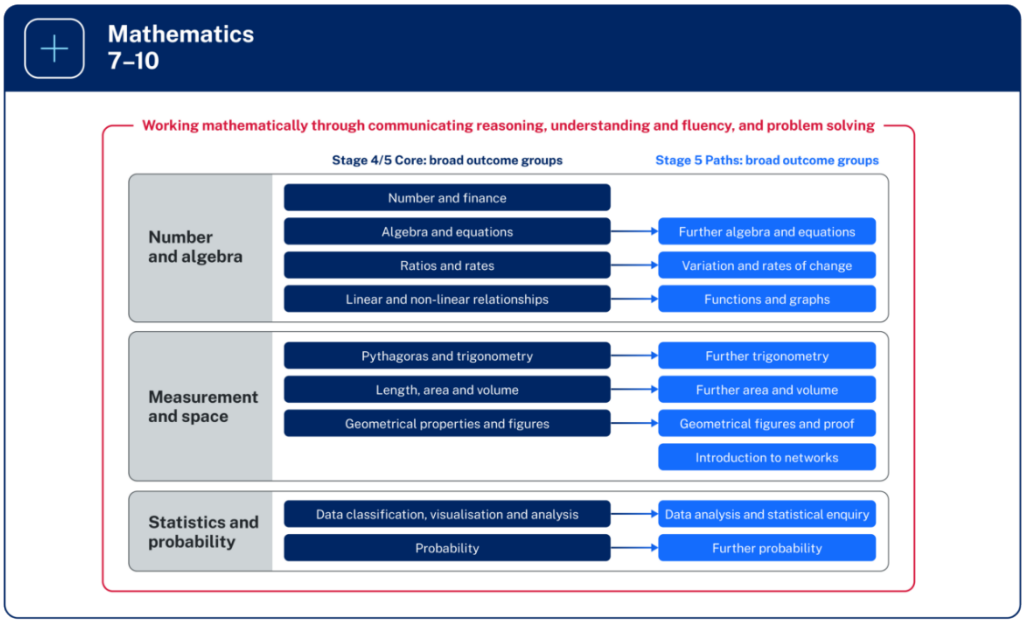


Figure 3: The organisation of Mathematics 7–10

Image long description: Stage 4/5 Core: broad outcome groups are Number and finance, Algebra and equations, Ratios and rates, Linear and non-linear relationships, Pythagoras and trigonometry, Length, area and volume, Geometrical properties and figures, Data classification, visualisation and analysis and Probability. Stage 5 Paths: broad outcome groups are Further algebra and equations, Variation and rates of change, Functions and graphs, Further trigonometry, Further area and

volume, Geometrical figures and proof, Introduction to networks, Data analysis and statistical enquiry and Further probability. All content is surrounded by the phrase, Working mathematically through communicating reasoning, understanding and fluency, and problem solving.

7–10 Core–Paths structure

The Core–Paths structure is designed to encourage aspiration in students. The structure is intended to extend students as far along the continuum of learning as possible and provide solid foundations for the highest levels of student achievement. The structure allows for a diverse range of endpoints up to the end of Stage 5.

Typically, the Core will cover teaching and learning experiences up to the middle of Stage 5. It is not the intention of the Core–Paths structure to lock students into predetermined pathways at the end of Stage 4.

Teachers program Path outcomes and content as appropriate to the needs of the students by the end of Stage 5. Paths may be implemented at any time in Stages 4 and 5.

The Core outcomes and content provide students with the foundation for Mathematics Standard 11–12.

Life Skills outcomes and content

Students with disability can access the syllabus outcomes and content in a range of ways. Decisions regarding curriculum options should be made in the context of collaborative curriculum planning.

Some students with intellectual disability may find the Years 7–10 Life Skills outcomes and content the most appropriate option to follow in Stage 4 and/or Stage 5. Before determining whether a student is eligible to undertake a course based on Life Skills outcomes and content, consideration should be given to other ways of assisting the student to engage with the Stage 4 and/or Stage 5 outcomes, or prior stage outcomes if appropriate. This assistance may include a range of adjustments to teaching, learning and assessment activities.

Life Skills outcomes cannot be taught in combination with other outcomes from the same subject. Teachers select specific Life Skills outcomes to teach based on the needs, strengths, goals, interests and prior learning of each student. Students are required to demonstrate achievement of one or more Life Skills outcomes.

Protocols for collaborating with Aboriginal and Torres Strait Islander Communities

NESA is committed to working in partnership with Aboriginal Communities and supporting teachers, schools and schooling sectors to improve educational outcomes for young people.

It is important to respect appropriate ways of interacting with Aboriginal Communities and Cultural material when teachers plan, program and implement learning experiences that focus on Aboriginal and Torres Strait Islander Priorities.

Indigenous Cultural and Intellectual Property (ICIP) protocols need to be followed. Aboriginal and Torres Strait Islander Peoples' ICIP protocols include Cultural Knowledges, Cultural Expression and Cultural Property and documentation of Aboriginal and Torres Strait Islander Peoples' identities and lived experiences. It is important to recognise the diversity and complexity of different Cultural groups in NSW, as protocols may differ between local Aboriginal Communities.

Teachers should work in partnership with Elders, parents, Community members, Cultural Knowledge Holders, or a local, regional or state Aboriginal Education Consultative Group. It is important to respect Elders and the roles of men and women. Local Aboriginal Peoples should be

invited to share their Cultural Knowledges with students and staff when engaging with Aboriginal histories and Cultural Practices.

Balance of content

The amount of content associated with a given outcome is not necessarily indicative of the amount of time spent engaging with the respective outcome. Teachers use formative and summative assessment to determine instructional priorities and the time needed for students to demonstrate expected outcomes.

In considering the intended learning, teachers make decisions about the sequence and emphasis to be given to particular groups of content based on the needs and abilities of their students.

Working at different stages

The content presented in a stage represents the typical knowledge, understanding and skills that students learn throughout the stage. It is acknowledged that students learn at different rates and in different ways. There may be students who will not demonstrate achievement in relation to one or more of the outcomes for the stage.

There may be instances where teachers will need to address outcomes across different stages in order to meet the learning needs of students. Teachers are best placed to make decisions about when students need to work at, above or below stage level in relation to one or more of the outcomes. This recognises that outcomes may be achieved by students at different times across stages. Only students who are accelerated in a course may access Stage 6 outcomes.

For example:

- Students in Early Stage 1 could be working on Stage 1 content in Number and algebra, while working on Early Stage 1 content in Measurement and space.
- In Stage 2 or Stage 3, some students may not have developed a complete understanding of place value and the role of zero to read, write and order two-digit and three-digit numbers. These students will need to access content from Early Stage 1 or Stage 1 before engaging with Stage 2 content in applying place value to larger numbers and decimals.
- In Stage 4 some students may not have developed a complete understanding of fractions, decimals and percentages and will need to access related outcomes from Stage 3.

Course description

Courses 7–10

The *Mathematics K–10 Syllabus* includes 2 courses for Mathematics 7–10:

- Mathematics
- Mathematics Life Skills

Focus of learning

Students learn how to demonstrate understanding of the relationships between mathematical concepts. They develop skills in solving routine and non-routine problems and learn to use mathematical language to solve problems and communicate reasoning.

What students learn

Students develop knowledge, understanding and skills across Number and algebra, Measurement and space, and Statistics and probability.

Through the Core, students develop knowledge of number, finance, algebraic expressions and equations, ratios and rates, linear and non-linear relationships, trigonometry, geometry, data and multistage probability.

Through the Paths, students extend knowledge of the core through more complex content, including further algebraic techniques, graphing, geometrical proof, networks, statistical enquiry and further probability.

Course requirements

Teachers program all core outcomes and content as part of students' curriculum entitlement.

Safety and risk management

Schools are required to ensure they follow safety and risk management in delivering the *Mathematics K–10 Syllabus*.

Life Skills

For Mathematics Life Skills:

- Students are required to demonstrate achievement of one or more Mathematics 7–10 Life Skills outcomes.
- Outcomes and content should be selected to meet the particular needs of individual students.
- The focus areas provide possible frameworks for addressing the Mathematics 7–10 Life Skills outcomes and content and are suggestions only. Teachers have the flexibility to select, group and sequence outcomes and content to meet the needs, strengths, goals, interests and prior learning of their students.
- The Working mathematically processes provide an opportunity for students to develop knowledge and skills in communicating reasoning, understanding and fluency, and problem solving. There is one Working mathematically outcome for all courses, including Mathematics 7–10 Life Skills. Teachers should embed Working mathematically processes into teaching and learning activities where appropriate to the needs and abilities of students.

Course enrolment details

Information about course entries for the RoSA are available on Assessment Certification Examination (ACE).

Mathematics – Stage 5

- Course number: 326
- Course hours: 200
- Enrolment type: Mandatory

Exclusions

- Mathematics Life Skills – Stage 5: 327

Students may not access Life Skills outcomes and other outcomes from the same subject.

Mathematics Life Skills – Stage 5

- Course number: 327
- Course hours: 200
- Enrolment type: Mandatory

Exclusions

- Mathematics – Stage 5: 326

Students may not access Life Skills outcomes and other outcomes from the same subject.

RoSA information

Information about curriculum requirements for the RoSA are available on Assessment Certification Examination (ACE).

Rationale

Mathematical ideas have evolved and continue to develop across cultures and have been practised in Australia by Aboriginal and Torres Strait Islander Peoples for thousands of years. Through the study of mathematics, students apply their knowledge and skills to deepen their understanding of the world.

Mathematics is a reasoning and creative activity, integral to scientific and technological advances across many fields of endeavour. The symbolic nature of mathematics provides a powerful and precise means of communication.

Making connections across mathematical concepts and other subject areas enhances students' ability to understand the purpose of learning mathematics and to develop a deeper conceptual understanding. This helps students to recognise the role of mathematics in solving problems in the world around them, applying their understanding to familiar and unfamiliar situations.

By studying mathematics, students develop essential numeracy skills and fluency, while nurturing the ability to think logically, critically and creatively. They learn about patterns and reason about relationships, creating opportunities to generalise their solutions and to solve non-routine problems.

When students enjoy learning mathematics, they develop a positive self-concept and become self-motivated learners through active participation in appropriately challenging tasks. This can enhance their resilience in solving mathematical problems relevant to further education and their everyday lives.

Aim

The aim of Mathematics K–10 is to enable students to become confident users of mathematics, learning and applying the language of mathematics to communicate efficiently and effectively. They develop an increasingly sophisticated understanding of mathematical concepts and a fluency with mathematical processes that helps them to interpret and solve problems. Students make connections within mathematics and connect mathematical concepts with the world around them. They learn to understand and appreciate how mathematics is a relevant part of their lives.

Table of outcomes

Secondary (7–10)

MAO-WM-01 Working mathematically

develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly

This outcome is aligned to all content in each Stage.

Focus area	Stage 4	Stage 5	Related Life Skills for Stages 4/5
Computation with integers	**MA4-INT-C-01** compares, orders and calculates with integers to solve problems	No Stage 5 outcomes	**MALS-LAN-01** recognises language that represents number **MALS-LAN-02** responds to and uses language that represents number **MALS-COU-01** counts in everyday contexts **MALS-REP-01** represents number in everyday contexts **MALS-COM-01** compares and orders numbers **MALS-ADS-01** uses strategies for addition and subtraction **MALS-MDI-01** uses strategies for multiplication and division

Focus area	Stage 4	Stage 5	Related Life Skills for Stages 4/5
Fractions, decimals and percentages	**MA4-FRC-C-01** represents and operates with fractions, decimals and percentages to solve problems	No Stage 5 outcomes	**MALS-FRC-01** demonstrates knowledge of fractions in everyday contexts **MALS-DEP-01** demonstrates knowledge of decimals and percentages in everyday contexts
Ratios and rates Includes: Stage 4: Ratios and rates Stage 5: Variation and rates of change A (Path) Stage 5: Variation and rates of change B (Path)	**MA4-RAT-C-01** solves problems involving ratios and rates, and analyses distance–time graphs	**MA5-RAT-P-01** identifies and solves problems involving direct and inverse variation and their graphical representations *(Path: Stn, Adv)* **MA5-RAT-P-02** analyses and constructs graphs relating to rates of change *(Path: Stn, Adv)*	**MALS-ADS-01** uses strategies for addition and subtraction **MALS-MDI-01** uses strategies for multiplication and division
Algebraic techniques Includes: Stage 4: Algebraic techniques Stage 5: Algebraic techniques A Stage 5: Algebraic techniques B (Path) Stage 5: Algebraic techniques C (Path)	**MA4-ALG-C-01** generalises number properties to operate with algebraic expressions including expansion and factorisation	**MA5-ALG-C-01** simplifies algebraic fractions with numerical denominators and expands algebraic expressions **MA5-ALG-P-01** simplifies algebraic fractions involving indices, and expands and factorises algebraic expressions *(Path: Adv)* **MA5-ALG-P-02** selects and applies appropriate algebraic techniques to operate with algebraic fractions, and expands, factorises and simplifies algebraic expressions *(Path: Adv)*	**MALS-PAT-01** recognises and applies patterns in everyday contexts

Focus area	Stage 4	Stage 5	Related Life Skills for Stages 4/5
Indices Includes: Stage 4: Indices Stage 5: Indices A Stage 5: Indices B (Path) Stage 5: Indices C (Path)	**MA4-IND-C-01** operates with primes and roots, positive-integer and zero indices involving numerical bases and establishes the relevant index laws	**MA5-IND-C-01** simplifies algebraic expressions involving positive-integer and zero indices, and establishes the meaning of negative indices for numerical bases **MA5-IND-P-01** applies the index laws to operate with algebraic expressions involving negative-integer indices *(Path: Adv)* **MA5-IND-P-02** describes and performs operations with surds and fractional indices *(Path: Adv)*	No Related Life Skills for Stages 4/5 outcomes
Equations Includes: Stage 4: Equations Stage 5: Equations A Stage 5: Equations B (Path) Stage 5: Equations C (Path)	**MA4-EQU-C-01** solves linear equations of up to 2 steps and quadratic equations of the form $ax^2 = c$	**MA5-EQU-C-01** solves linear equations of up to 3 steps, limited to one algebraic fraction **MA5-EQU-P-01** solves monic quadratic equations, linear inequalities and cubic equations of the form $ax^3 = k$ *(Path: Adv)* **MA5-EQU-P-02** solves linear equations of more than 3 steps, monic and non-monic quadratic equations, and linear simultaneous equations *(Path: Adv)*	**MALS-ADS-01** uses strategies for addition and subtraction **MALS-MDI-01** uses strategies for multiplication and division
Linear relationships	**MA4-LIN-C-01**	**MA5-LIN-C-01**	**MALS-POS-01**

Focus area	Stage 4	Stage 5	Related Life Skills for Stages 4/5
Includes: Stage 4: Linear relationships Stage 5: Linear relationships A Stage 5: Linear relationships B Stage 5: Linear relationships C (Path)	creates and displays number patterns and finds graphical solutions to problems involving linear relationships	determines the midpoint, gradient and length of an interval, and graphs linear relationships, with and without digital tools **MA5-LIN-C-02** graphs and interprets linear relationships using the gradient/slope-intercept form **MA5-LIN-P-01** describes and applies transformations, the midpoint, gradient/slope and distance formulas, and equations of lines to solve problems *(Path: Adv)*	demonstrates knowledge of position and direction in everyday contexts
Length	**MA4-LEN-C-01** applies knowledge of the perimeter of plane shapes and the circumference of circles to solve problems	No Stage 5 outcomes	**MALS-LEN-01** measures and uses length in everyday contexts
Pythagoras and trigonometry Includes: Stage 4: Right-angled triangles (Pythagoras' theorem) Stage 5: Trigonometry A Stage 5: Trigonometry B Stage 5: Trigonometry C (Path) Stage 5: Trigonometry D (Path)	**MA4-PYT-C-01** applies Pythagoras' theorem to solve problems in various contexts	**MA5-TRG-C-01** applies trigonometric ratios to solve right-angled triangle problems **MA5-TRG-C-02** applies trigonometry to solve problems, including bearings and angles of elevation and depression **MA5-TRG-P-01** applies Pythagoras' theorem and trigonometry to solve 3-dimensional problems and applies	No Related Life Skills for Stages 4/5 outcomes

Focus area	Stage 4	Stage 5	Related Life Skills for Stages 4/5
		the sine, cosine and area rules to solve 2-dimensional problems, including bearings *(Path: Stn, Adv)* **MA5-TRG-P-02** establishes and applies the properties of trigonometric functions and finds solutions to trigonometric equations *(Path: Adv)*	
Area and surface area Includes: Stage 4: Area Stage 5: Area and surface area A Stage 5: Area and surface area B (Path)	**MA4-ARE-C-01** applies knowledge of area and composite area involving triangles, quadrilaterals and circles to solve problems	**MA5-ARE-C-01** solves problems involving the surface area of right prisms and practical problems involving the area of composite shapes and solids **MA5-ARE-P-01** applies knowledge of the surface area of right pyramids and cones, spheres and composite solids to solve problems *(Path: Stn, Adv)*	**MALS-ARE-01** measures and uses area in everyday contexts
Volume Includes: Stage 4: Volume Stage 5: Volume A Stage 5: Volume B (Path)	**MA4-VOL-C-01** applies knowledge of volume and capacity to solve problems involving right prisms and cylinders	**MA5-VOL-C-01** solves problems involving the volume of composite solids consisting of right prisms and cylinders **MA5-VOL-P-01** applies knowledge of the volume of right pyramids, cones and spheres to solve problems involving related composite solids *(Path: Stn, Adv)*	**MALS-VOL-01** measures and uses volume, capacity and mass in everyday contexts
Angle relationships	**MA4-ANG-C-01**	No Stage 5 outcomes	No Related Life Skills for Stages

Focus area	Stage 4	Stage 5	Related Life Skills for Stages 4/5
	applies angle relationships to solve problems, including those related to transversals on sets of parallel lines		4/5 outcomes
Properties of geometrical figures Includes: Stage 4: Properties of geometrical figures Stage 5: Properties of geometrical figures A Stage 5: Properties of geometrical figures B (Path) Stage 5: Properties of geometrical figures C (Path)	**MA4-GEO-C-01** identifies and applies the properties of triangles and quadrilaterals to solve problems	**MA5-GEO-C-01** identifies and applies the properties of similar figures and scale drawings to solve problems **MA5-GEO-P-01** establishes conditions for congruent triangles and similar triangles and solves problems relating to properties of similar figures and plane shapes *(Path: Ext)* **MA5-GEO-P-02** constructs proofs involving congruent triangles and similar triangles and proves properties of plane shapes *(Path: Ext)*	**MALS-GEO-01** explores 2-dimensional shapes and 3-dimensional objects
Data classification and visualisation	**MA4-DAT-C-01** classifies and displays data using a variety of graphical representations	No Stage 5 outcomes	**MALS-DAT-01** recognises and represents data in everyday contexts
Data analysis Includes: Stage 4: Data analysis Stage 5: Data analysis A Stage 5: Data analysis B Stage 5: Data analysis C (Path)	**MA4-DAT-C-02** analyses simple datasets using measures of centre, range and shape of the data	**MA5-DAT-C-01** compares and analyses datasets using summary statistics and graphical representations **MA5-DAT-C-02** displays and interprets datasets involving bivariate data	**MALS-DAT-02** interprets information from data displays

Focus area	Stage 4	Stage 5	Related Life Skills for Stages 4/5
		MA5-DAT-P-01 plans, conducts and reviews a statistical inquiry into a question of interest *(Path: Stn, Adv)*	
Probability Includes: Stage 4: Probability Stage 5: Probability A Stage 5: Probability B (Path)	**MA4-PRO-C-01** solves problems involving the probabilities of simple chance experiments	**MA5-PRO-C-01** solves problems involving probabilities in multistage chance experiments and simulations **MA5-PRO-P-01** solves problems involving Venn diagrams, 2-way tables and conditional probability *(Path: Adv)*	**MALS-PRO-01** applies chance and probability to everyday events
Financial mathematics Includes: Stage 5: Financial mathematics A Stage 5: Financial mathematics B	No Stage 4 outcomes	**MA5-FIN-C-01** solves financial problems involving simple interest, earning money and spending money **MA5-FIN-C-02** solves financial problems involving compound interest and depreciation	**MALS-FIN-01** demonstrates knowledge of money in everyday contexts **MALS-FIN-02** plans and manages personal finances
Non-linear relationships Includes: Stage 5: Non-linear relationships A Stage 5: Non-linear relationships B Stage 5: Non-linear relationships C (Path)	No Stage 4 outcomes	**MA5-NLI-C-01** identifies connections between algebraic and graphical representations of quadratic and exponential relationships in various contexts **MA5-NLI-C-02** identifies and compares features of parabolas and exponential curves in various contexts	No Related Life Skills for Stages 4/5 outcomes

Focus area	Stage 4	Stage 5	Related Life Skills for Stages 4/5
		MA5-NLI-P-01 interprets and compares non-linear relationships and their transformations, both algebraically and graphically *(Path: Adv)*	
Numbers of any magnitude	No Stage 4 outcomes	**MA5-MAG-C-01** solves measurement problems by using scientific notation to represent numbers and rounding to a given number of significant figures	No Related Life Skills for Stages 4/5 outcomes
Polynomials (Path)	No Stage 4 outcomes	**MA5-POL-P-01** defines, operates with and graphs polynomials and applies the factor and remainder theorems to solve problems *(Path: Adv, Ext)*	No Related Life Skills for Stages 4/5 outcomes
Logarithms (Path)	No Stage 4 outcomes	**MA5-LOG-P-01** establishes and applies the laws of logarithms to solve problems *(Path: Adv)*	No Related Life Skills for Stages 4/5 outcomes
Functions and other graphs (Path)	No Stage 4 outcomes	**MA5-FNC-P-01** uses function notation to describe and graph functions of one variable and graphs inequalities in one and 2 variables *(Path: Adv)*	No Related Life Skills for Stages 4/5 outcomes
Circle geometry (Path)	No Stage 4 outcomes	**MA5-CIR-P-01** applies deductive reasoning to prove circle theorems and solve related problems *(Path: Ext)*	No Related Life Skills for Stages 4/5 outcomes
Introduction to networks (Path)	No Stage 4 outcomes	**MA5-NET-P-01**	No Related Life Skills for Stages

Focus area	Stage 4	Stage 5	Related Life Skills for Stages 4/5
		solves problems involving the characteristics of graphs/networks, planar graphs and Eulerian trails and circuits *(Path: Stn)*	4/5 outcomes

Before deciding that a student should undertake a course based on Life Skills outcomes and content, consideration should be given to other ways of assisting the student to engage with the Stage 4 or Stage 5 outcomes. Further information in relation to planning, implementing and assessing Life Skills outcomes and content can be found on the NESA website.

Outcomes and content for Stage 4

Computation with integers

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- compares, orders and calculates with integers to solve problems **MA4-INT-C-01**

Related Life Skills outcomes: MALS-LAN-01, MALS-LAN-02, MALS-COU-01, MALS-REP-01, MALS-COM-01, MALS-ADS-01, MALS-MDI-01

Content

Compare and order integers

- Recognise and describe the direction and magnitude of integers
- Identify and represent integers on a number line
- Compare the relative value of integers using the less than (<) and greater than (>) symbols
- Order integers

Add and subtract positive and negative integers

- Add and subtract integers with and without the use of digital tools
- Construct a directed number sentence to model a situation
- Examine different meanings (position or operation) for the $+$ and $-$ signs, depending on context

Multiply and divide positive and negative integers

- Represent multiples of negative integers as repeated addition
- Multiply and divide positive and negative integers with and without the use of digital tools

Apply the 4 operations to integers

- Apply the 4 operations to integers
- Solve problems involving grouping symbols with integers
- Apply the order of operations to evaluate expressions involving integers, with and without the use of digital tools

Fractions, decimals and percentages

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- represents and operates with fractions, decimals and percentages to solve problems **MA4-FRC-C-01**

Related Life Skills outcomes: MALS-FRC-01, MALS-DEP-01

Content

Compare fractions using equivalence

- Determine the highest common factor (HCF) of 2 whole numbers
- Examine methods of generating equivalent fractions
- Simplify fractions by using methods, including determining the HCF of the numerator and denominator or repeated simplification using common factors
- Create fractions with the same denominator to compare their sizes
- Compare and order fractions with different denominators

Round decimals to a specified degree of accuracy using approximations

- Round decimals to a given number of decimal places
- Apply the notation ≈ as a symbol of numerical approximation
- Reason why an approximation may be more appropriate than an exact answer and vice versa

Identify terminating and recurring decimals

- Use either dot or vinculum notation for recurring (repeating) decimals
- Classify decimals as recurring or terminating

Identify and make use of the relationship between fractions, decimals and percentages to carry out simple conversions

- Define rational numbers as numbers that can be written in the form $\frac{a}{b}$, where a and b are integers and $b \neq 0$
- Classify fractions and percentages as rational numbers
- Recognise and explain that numbers with terminating or recurring decimals are rational
- Represent fractions as decimals (terminating and recurring) and percentages
- Represent terminating decimals as fractions and percentages
- Represent improper fractions as mixed numbers and decimals, and vice versa
- Represent percentages as fractions and decimals

Examine the concept of irrational numbers

- Identify and define irrational numbers as numbers that cannot be written in the form $\frac{a}{b}$ where a and b are integers and $b \neq 0$
- Find approximations of irrational numbers using digital tools
- Locate the approximate position of irrational numbers on a number line

Order and compare the value of fractions, decimals and percentages

- Locate positive and negative fractions, decimals and mixed numbers on a number line to compare their relative values
- Compare and order fractions, mixed numbers, decimals (terminating and recurring) and percentages

Solve problems that involve the addition and subtraction of fractions

- Represent addition and subtraction of fractions with the same or unrelated denominators
- Solve problems involving adding and subtracting fractions and mixed numbers, including finding a common denominator
- Solve problems that involve subtracting a fraction from a whole number, with and without the use of digital tools

Solve problems that involve the multiplication and division of fractions and decimals

- Compare and generalise the effect of multiplying or dividing by a number with magnitude between zero and one
- Represent multiplication and division of decimals
- Represent multiplication and division of fractions, including mixed numbers
- Multiply and divide decimals, using digital tools to solve problems
- Multiply and divide fractions and mixed numbers, with and without using digital tools to solve problems
- Compare initial estimates with the results of calculations
- Apply knowledge of fractions and decimals of quantities to solve problems
- Apply knowledge of multiplication and division of fractions and decimals to solve problems

Represent one quantity as a fraction, decimal or percentage of another, with and without the use of digital tools

- Represent one quantity as a fraction, decimal or percentage of another by considering appropriate units
- Calculate percentage increases and decreases in various contexts
- Examine the financial applications of percentage increase and decrease, including profit and/or loss as a percentage of cost price or selling price

Solve problems that involve the use of percentages

- Apply knowledge of percentages to calculate quantities in various contexts
- Apply knowledge of percentage increases and decreases to solve problems in various contexts
- Solve real-life problems involving percentages using the unitary method or other techniques
- Solve financial problems involving percentages, specifically considering GST, profit and loss

Ratios and rates

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- solves problems involving ratios and rates, and analyses distance–time graphs **MA4-RAT-C-01**

Related Life Skills outcomes: MALS-ADS-01, MALS-MDI-01

Content

Recognise and simplify ratios

- Use ratios to compare 2 or more quantities measured in the same units
- Identify and express one part of a ratio as a fraction of the whole
- Simplify ratios

Solve problems involving ratios

- Apply the unitary method to solve ratio problems
- Divide a quantity in a given ratio
- Solve real-life problems involving ratios

Recognise and simplify rates

- Explain the differences between ratios and rates
- Represent given information as a simplified rate
- Convert between units for rates

Solve problems involving rates

- Solve a variety of real-life problems involving rates
- Examine financial applications of rates, such as best buys

Interpret and construct distance–time graphs from authentic data

- Interpret distance–time graphs made up of straight-line segments with a negative, zero or positive slope
- Calculate speeds for straight-line segments of given distance–time graphs
- Create distance–time graphs made up of straight-line segments

Algebraic techniques

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- generalises number properties to operate with algebraic expressions including expansion and factorisation **MA4-ALG-C-01**

Related Life Skills outcomes: MALS-PAT-01

Content

Examine the concept of pronumerals as a way of representing numbers

- Examine and recognise that pronumerals can be used to represent one or more numerical values and when pronumerals have more than one numerical value, they may then be referred to as variables
- Identify and define an algebraic expression as an expression formed by combining numbers and algebraic symbols using arithmetic operations
- Use concise algebraic notation and conventions for multiplication, division and powers, and explain the meanings for each convention

Create algebraic expressions and evaluate them by substitution

- Generate algebraic expressions by translating descriptions and vice versa
- Substitute numbers into algebraic expressions and evaluate the result
- Generate a number pattern from an algebraic expression

Extend and apply the laws and properties of arithmetic to algebraic terms and expressions

- Generalise the associative property of addition and multiplication to algebraic expressions
- Generalise the commutative property to algebraic expressions
- Identify like terms, and add and subtract them to simplify algebraic expressions
- Simplify algebraic expressions that involve multiplication and division, including simple algebraic fractions
- Simplify algebraic expressions involving mixed operations

Extend and apply the distributive law to the expansion of algebraic expressions

- Explain the role and meaning of grouping symbols in algebraic expressions
- Apply the distributive law to expand and simplify algebraic expressions by removing grouping symbols

Factorise algebraic expressions by identifying numerical and algebraic factors

- Identify and list factors of a single term
- Factorise algebraic expressions using knowledge of factors and finding the highest common numerical factor (HCF)
- Factorise algebraic expressions using knowledge of factors by finding a common algebraic factor, including expressions involving more than 2 terms, and verify the result by expansion

Indices

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- operates with primes and roots, positive-integer and zero indices involving numerical bases and establishes the relevant index laws **MA4-IND-C-01**

Content

Apply index notation to represent whole numbers as products of powers of prime numbers

- Describe numbers written in index form using terms such as *base*, *power*, *index* and *exponent*
- Represent numbers in index notation limited to positive powers
- Represent in expanded form and evaluate numbers expressed in index notation, including powers of 10
- Apply the order of operations to evaluate expressions involving indices
- Determine and apply tests for divisibility for 2, 3, 4, 5, 6 and 10
- Represent a whole number greater than one as a product of its prime factors, using index notation where appropriate

Examine cube roots and square roots

- Use the notations for square root ($\sqrt{\ }$) and cube root ($\sqrt[3]{\ }$)
- Recognise and describe the relationship between squares and square roots, and cubes and cube roots for positive numbers
- Verify, through numerical examples, that $\sqrt{ab} = \sqrt{a} \times \sqrt{b}$
- Estimate the square root of any non-square whole number and the cube root of any non-cube whole number, then check using a calculator
- Identify and describe exact and approximate solutions in the context of square roots and cube roots
- Apply the order of operations to evaluate expressions involving square roots, cube roots, square numbers and cube numbers

Use index notation to establish the index laws with positive-integer indices and the zero index

- Establish the multiplication, division and the power of a power index laws, by expressing each number in expanded form with numerical bases and positive-integer indices
- Verify through numerical examples that $(ab)^2 = a^2b^2$
- Establish the meaning of the zero index
- Apply index laws to simplify and evaluate expressions with numerical bases

Equations

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- solves linear equations of up to 2 steps and quadratic equations of the form $ax^2 = c$ **MA4-EQU-C-01**

Related Life Skills outcomes: MALS-ADS-01, MALS-MDI-01

Content

Solve linear equations up to 2 steps

- Represent number sentences involving unknown quantities using pronumerals
- Describe number sentences as equations
- Distinguish between and compare algebraic expressions and equations
- Solve linear equations with integer and non-integer solutions using algebraic techniques that involve up to 2 steps, including equations with pronumerals on both sides
- Model and solve word problems using equations of up to 2 steps

Solve and verify linear equations by substitution

- Verify solutions to equations by substitution
- Solve problems involving linear equations, including those arising from substituting given values into formulas

Solve quadratic equations

- Reason why there are 2 values of x that satisfy a quadratic equation of the form $x^2 = c$ if $c > 0$
- Solve problems involving quadratic equations of the form $ax^2 = c$, giving answers in exact form and as decimal approximations
- Solve quadratic equations arising from substitution into a formula

Linear relationships

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- creates and displays number patterns and finds graphical solutions to problems involving linear relationships **MA4-LIN-C-01**

Related Life Skills outcomes: MALS-POS-01

Content

Plot and identify points on the Cartesian plane

- Plot and label points on the Cartesian plane of given coordinates, including those with coordinates that are not whole numbers
- Identify and record the coordinates of given points on the Cartesian plane, including those with coordinates that are not whole numbers

Plot linear relationships on the Cartesian plane

- Construct a geometric pattern and record the results in a table of values
- Represent a given number pattern (including decreasing patterns) using a table of values
- Describe a number pattern in words and generate an equation using algebraic symbols
- Apply an equation generated from a pattern to calculate the corresponding value for a smaller or larger number
- Recognise that a linear relationship can be represented by a number pattern, an equation (or a rule using algebraic symbols), a table of values, a set of pairs of coordinates and a line graphed on a Cartesian plane, and move flexibly between these representations
- Explain that there are an infinite number of ordered pairs that satisfy a given linear relationship by extending a line joining a set of points on the Cartesian plane
- Compare similarities and differences of multiple straight-line graphs on the same set of axes using graphing applications
- Describe linear relationships in real-life contexts and solve related problems

Solve linear equations using graphical techniques

- Recognise that each point on the graph of a linear relationship satisfies the equation of a line
- Apply graphs of linear relationships to solve a corresponding linear equation using graphing applications
- Graph 2 intersecting lines on the same set of axes and identify the point of intersection using either graphing applications or a table of values
- Verify that the point of intersection satisfies the equations of both lines

Length

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- applies knowledge of the perimeter of plane shapes and the circumference of circles to solve problems **MA4-LEN-C-01**

Related Life Skills outcomes: MALS-LEN-01

Content

Solve problems involving the perimeter of various quadrilaterals and simple composite figures

- Solve problems involving the perimeter of plane shapes, including parallelograms, trapeziums, rhombuses and kites
- Solve problems relating to the perimeter of simple composite figures
- Compare methods of solution for finding perimeter and evaluate the efficiency of those methods

Describe the relationships between the features of circles

- Identify and describe the relationship between circle features, including the radius, diameter, arc, chord, sector and segment of a circle, and a tangent to a circle
- Define π as the ratio of the circumference to the diameter of any circle
- Verify that the number π is a constant and develop the formula for the circumference of a circle
- Apply the formula for the circumference of a circle in terms of the diameter d or radius r (circumference of a circle $= \pi d$ or $2\pi r$) to solve related problems
- Establish the arc length formula $\left(l = \frac{\theta}{360} \times 2\pi r\right)$ where l is the arc length and θ is the angle subtended at the centre by the arc
- Solve problems by finding arc lengths and the perimeter of sectors, giving an exact answer in terms of π or an approximate answer
- Find the perimeter of quadrants, semicircles and simple composite figures consisting of 2 shapes in a variety of contexts, including using digital tools

Right-angled triangles (Pythagoras' theorem)

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- applies Pythagoras' theorem to solve problems in various contexts **MA4-PYT-C-01**

Content

Identify and define Pythagoras' theorem

- Identify and describe the hypotenuse as the side opposite the right angle and the longest side in any right-angled triangle
- Establish the relationship between the lengths of the sides of a right-angled triangle
- Use the relationship to record and define Pythagoras' theorem both algebraically and in words

Examine problems involving Pythagoras' theorem

- Apply Pythagoras' theorem to find the unknown length of a side in a right-angled triangle, giving answers in an exact form or as decimal approximations
- Apply the converse of Pythagoras' theorem to establish whether a triangle is right angled
- Solve practical problems involving Pythagoras' theorem before exploring a variety of related problems
- Justify whether a set of 3 integers is a Pythagorean triad

Area

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- applies knowledge of area and composite area involving triangles, quadrilaterals and circles to solve problems **MA4-ARE-C-01**

Related Life Skills outcomes: MALS-ARE-01

Content

Develop and use formulas to find the area of rectangles, triangles and parallelograms to solve problems

- Apply the formula to find the area of a rectangle or square: $A = lb$, where l is the length and b is the breadth (or width) of the rectangle or square
- Develop and apply the formula to find the area of a triangle: $A = \frac{1}{2}bh$, where b is the base length and h is the perpendicular height
- Develop and apply the formula to find the area of a parallelogram: $A = bh$ where b is the base length and h is the perpendicular height
- Calculate the area of composite figures that can be dissected into rectangles, squares, parallelograms or triangles to solve problems

Develop and use the formula to find the area of circles and sectors to solve problems

- Develop and apply the formula to find the area of a circle: $A = \pi r^2$, where r is the length of the radius
- Explain how the area of a sector can be developed from the area of a circle $\left(A = \frac{\theta}{360} \times \pi r^2\right)$
- Find the area of quadrants, semicircles and sectors, and apply these formulas in the context of real-life problems
- Calculate the areas of composite shapes involving quadrants, semicircles and sectors to solve problems

Develop and use the formulas to find the area of trapeziums, rhombuses and kites to solve problems

- Develop and apply the formula to find the area of a kite or rhombus: $A = \frac{1}{2}xy$, where x and y are the lengths of the diagonals
- Develop and apply the formula to find the area of a trapezium: $A = \frac{h}{2}(a + b)$, where h is the perpendicular height and a and b are the lengths of parallel sides
- Calculate the area of composite shapes involving trapeziums, kites and rhombuses to solve problems

Choose appropriate units of measurement for area and convert between units

- Choose an appropriate unit to measure the area of different shapes and surfaces, and justify the choice

- Convert between metric units of area using 1 cm^2 $=$ 100 mm^2, 1 m^2 $=$ $10\,000$ cm^2, 1 ha $=$ $10\,000$ m^2 and 1 km^2 $=$ $1\,000\,000$ m^2 $=$ 100 ha

Volume

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- applies knowledge of volume and capacity to solve problems involving right prisms and cylinders **MA4-VOL-C-01**

Related Life Skills outcomes: MALS-VOL-01

Content

Describe the different views of prisms and solids that have been formed from prism combinations

- Represent prisms from different views in 2 dimensions, including top, side, front and back views
- Describe and illustrate solids formed from prism combinations from different views in 2 dimensions, including top, side, front and back views
- Identify and illustrate the cross-sections of different prisms
- Examine the idea that prisms have a uniform cross-section that is equal to the base area
- Determine if a particular solid has a uniform cross-section

Develop and apply the formula to find the volume of a prism to solve problems

- Develop the formula for the volume of a prism: $V = \textit{base area} \times \textit{height}$, leading to $V = Ah$
- Apply the formula for the volume of a prism to prisms with uniform cross-sections to solve problems

Develop the formula for finding the volume of a cylinder and apply the formula to solve problems

- Develop and apply the formula to solve problems involving the volume of cylinders: $V = \pi r^2 h$, where r is the length of the radius of the base and h is the perpendicular height

Choose appropriate units of measurement for volume and capacity and convert between units

- Recognise that 1000 L is equal to 1 kilolitre (kL) and use the abbreviation
- Recognise that 1000 kL is equal to 1 megalitre (ML) and use the abbreviation
- Choose an appropriate unit to measure the volume or capacity of different objects and justify the choice
- Convert between metric units of volume and capacity (1 cm^3 = 1000 mm^3, 1 cm^3 = 1 mL, 1 m^3 = 1000 L = 1 kL, 1000 kL = 1 ML)
- Solve practical problems involving the volume and capacity of right prisms and cylinders

Angle relationships

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- applies angle relationships to solve problems, including those related to transversals on sets of parallel lines **MA4-ANG-C-01**

Content

Apply the language, notation and conventions of geometry

- Use appropriate terminology and conventions to define, label and name points, rays, lines and intervals using capital letters
- Identify and label the vertex and arms of an angle with capital letters
- Use appropriate conventions to label and name angles
- Use common conventions to indicate right angles, equal angles and intervals on diagrams

Identify geometrical properties of angles at a point

- Identify right angles, straight angles, angles of complete revolution and vertically opposite angles
- Apply the terms *complementary* and *supplementary* to a pair of angles adding to $90°$and $180°$, respectively
- Apply the term *adjacent angles* to a pair of angles with a common arm and common vertex

Identify and describe corresponding, alternate and co-interior angles when 2 straight lines are crossed by a transversal, including parallel lines

- Identify and describe perpendicular lines using the symbol for *is perpendicular to* ($\perp$)
- Apply the common conventions to indicate parallel lines on diagrams
- Identify and describe pairs of parallel lines using the symbol for *is parallel to* ($\parallel$)
- Identify and define transversals, including transversals of parallel lines
- Identify, name and measure alternate angle pairs, corresponding angle pairs and co-interior angle pairs for 2 lines cut by a transversal
- Verify and identify corresponding angles and alternate angles as equal, and co-interior angles as supplementary, when a pair of parallel lines is cut by a transversal
- Justify that 2 lines are parallel by using properties of alternate, corresponding or co-interior angles on parallel lines

Solve numerical problems involving angles using reasoning

- Apply the knowledge of angle relationships including angles at a point to find the sizes of unknown angles embedded in diagrams and give reasons
- Apply the knowledge of angles associated with parallel lines to find the sizes of unknown angles embedded in related diagrams and give reasons

Properties of geometrical figures

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- identifies and applies the properties of triangles and quadrilaterals to solve problems **MA4-GEO-C-01**

Related Life Skills outcomes: MALS-GEO-01

Content

Classify triangles according to their side and angle properties

- Label triangles using appropriate text and symbols
- Classify and describe types of triangles based on their properties, including acute-angled, right-angled, obtuse-angled, equilateral, isosceles and scalene triangles

Classify quadrilaterals and describe their properties

- Identify quadrilaterals using naming conventions
- Distinguish between convex and non-convex quadrilaterals
- Verify and describe the properties of the special quadrilaterals which include parallelograms, rectangles, rhombuses, squares, trapeziums and kites
- Identify and label the properties of the special quadrilaterals using appropriate conventions
- Classify quadrilaterals based on their properties
- Justify why some quadrilaterals may be classified as more than one type of quadrilateral

Apply the properties of triangles and quadrilaterals

- Prove that the interior angle sum of a triangle is 180° with or without digital tools
- Prove that any exterior angle of a triangle equals the sum of the 2 interior opposite angles
- Apply the angle sum of a triangle to prove that the angle sum of a quadrilateral is 360°
- Apply the properties of triangles and quadrilaterals to determine unknown sides and angles to solve problems, giving reasons

Data classification and visualisation

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- classifies and displays data using a variety of graphical representations **MA4-DAT-C-01**

Related Life Skills outcomes: MALS-DAT-01

Content

Classify data as either numerical (discrete or continuous) or categorical (nominal or ordinal) variables

- Define a variable in the context of statistics as any characteristic, number or quantity that can be measured or counted
- Classify and describe variables as numerical or categorical
- Describe a numerical variable as either discrete or continuous
- Describe a categorical variable as nominal or ordinal
- Distinguish between and compare numerical (discrete or continuous) and categorical (nominal or ordinal) variables

Display data using graphical representations relevant to the purpose of the data

- Represent single datasets using graphs, including frequency histograms and polygons, dot plots, stem-and-leaf plots, divided bar graphs, column graphs, line graphs, sector graphs and pictograms, with or without digital tools
- Include sources, titles, labels and scales when displaying data in a graph
- Select the type of graph best suited to represent various single datasets and justify the choice of graph
- Represent a dataset using a statistical infographic and justify the choice of graphical representation used

Interpret data in graphical representations

- Identify and interpret data displayed on graphs
- Identify features of graphical representations to draw conclusions
- Interpret patterns in graphical representations to make predictions
- Explain why a given graphical representation can lead to a misinterpretation of data

Data analysis

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- analyses simple datasets using measures of centre, range and shape of the data **MA4-DAT-C-02**

Related Life Skills outcomes: MALS-DAT-02

Content

Calculate and compare the mean, median, mode and range for simple datasets

- Calculate the mean *(x)* of a set of data using digital tools
- Calculate and describe the mean, median, mode and range of a dataset
- Classify the mean, median and/or mode as measure(s) of centre to represent the average or typical value of a dataset
- Describe and interpret data displays using mean, median and range
- Identify and describe datasets as having no modes (uniform), one mode (unimodal), 2 modes (bimodal) or multiple modes (multimodal)
- Identify the range as a measure of spread to describe variation in a dataset
- Compare simple datasets using the mean, median, mode and range

Interpret the effect individual data points have on measures of centre and range

- Informally identify clusters, gaps and outliers in datasets and give reasons for their occurrence in the context of the data
- Identify and explain the impact of adding or removing data values that are clustered at one end of a dataset on the measures of centre
- Identify and explain the impact of outliers on the measures of centre and range
- Determine and justify the most appropriate measure of centre to summarise the data in its context

Analyse datasets presented in various ways and draw conclusions

- Identify and describe the shape and distribution of a dataset using the terms *symmetrical*, *negatively skewed* and *positively skewed*
- Define a census as a study of every unit, everyone or everything in a population
- Define a sample as a subset of units in a population selected to represent all units in a population of interest
- Draw conclusions and make informed decisions about data gathered using data-collection techniques, including census and sampling, which is then presented in tables, graphs and charts

Probability

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- solves problems involving the probabilities of simple chance experiments **MA4-PRO-C-01**

Related Life Skills outcomes: MALS-PRO-01

Content

Determine probabilities for chance experiments

- List the sample space for chance experiments
- Express the probability of an event, which has a finite number of equally likely outcomes, as $P(\text{event}) = \frac{\text{number of favourable outcomes}}{\text{total number of outcomes}}$
- Recognise that probabilities range from 0 (impossible) to 1 (certain) and that equally likely outcomes have equal probabilities
- Verify that the total of the probabilities of all possible outcomes of an event is 1
- Identify and describe theoretical (expected) probabilities as being the likelihood of outcomes occurring under fair or unbiased conditions
- Explain that observed probability is the relative frequency resulting from repeated trials of a simulation and determine observed probabilities
- Explore relative frequencies by using a random number generator to repeat a chance experiment a number of times

Determine probabilities for complementary events

- Identify and describe the complement of an event
- Verify that the sum of the probability of an event and its complement is a total of 1
- Solve problems involving the probability of complementary events
- Represent the possible outcomes for complementary events in various forms

Outcomes and content for Stage 5

Financial mathematics A

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- solves financial problems involving simple interest, earning money and spending money **MA5-FIN-C-01**

Related Life Skills outcomes: MALS-FIN-01, MALS-FIN-02

Content

Solve problems involving earning money

- Solve problems involving wages given an hourly rate of pay including penalty rates for overtime, weekends and public holidays
- Calculate earnings from non-wage sources exploring commission, piecework and royalties
- Calculate weekly, fortnightly, monthly and yearly earnings assuming 1 year = 52 weeks
- Calculate leave loading by finding a percentage of eligible normal pay
- Investigate sources of published tables or online calculators and use these to calculate the weekly, fortnightly or monthly tax to be deducted from a worker's pay under the Australian Pay-As-You-Go (PAYG) taxation system
- Determine annual taxable income by exploring allowable deductions and current tax rates
- Calculate net earnings after deductions and taxation

Solve problems involving simple interest

- Establish and use the formula $I = Prn$ to find simple interest where $I =$ simple interest, $P =$ principal, $r =$ interest rate per time period and $n =$ number of time periods
- Apply the simple interest formula to solve problems related to investing money at simple interest rates, both algebraically and graphically

Solve problems involving spending money

- Calculate the cost of buying items on terms, by paying an initial deposit and making regular repayments
- Examine payment options involving *buy now*, *pay later* and investigate the costs associated with these schemes for purchasing goods
- Examine the principles behind short-term loans involving small dollar amounts and compare borrowing costs associated with using these products

Financial mathematics B

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- solves financial problems involving compound interest and depreciation **MA5-FIN-C-02**

Related Life Skills outcomes: MALS-FIN-01, MALS-FIN-02

Content

Solve problems involving compound interest and depreciation

- Examine compound interest for up to 3 time periods using repetition of the formula for simple interest
- Associate the calculation of the total value of a compound interest investment with repeated multiplication, using digital tools
- Establish and use the formula $FV = PV(1 + r)^n$ to find compound interest where $FV =$ future value of the investment, $PV =$ present value of the investment, $r =$ interest rate per time period and $n =$ number of time periods
- Solve problems involving compound interest
- Compare simple interest with compound interest in practical situations
- Use the compound interest formula to establish the depreciation formula $S = V_0(1 - r)^n$ where $S =$ salvage value, $V_0 =$ initial value of the asset, $r =$ depreciation rate per time period and $n =$ number of periods
- Solve problems involving the depreciation of an asset

Algebraic techniques A

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- simplifies algebraic fractions with numerical denominators and expands algebraic expressions **MA5-ALG-C-01**

Related Life Skills outcomes: MALS-PAT-01

Content

Apply the 4 operations to simplify algebraic fractions with numerical denominators

- Simplify expressions that involve algebraic fractions with numerical denominators

Apply the distributive law to the expansion of algebraic expressions, and collect like terms where appropriate

- Expand algebraic expressions, including those with negative coefficients
- Expand and simplify algebraic expressions by removing grouping symbols and collecting like terms where appropriate
- Expand binomial products algebraically using the distributive law or consider areas of rectangles as a possible method of expansion

Indices A

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- simplifies algebraic expressions involving positive-integer and zero indices, and establishes the meaning of negative indices for numerical bases **MA5-IND-C-01**

Content

Extend and apply the index laws to variables, using positive-integer indices and the zero index

- Apply the index laws for numerical bases with positive-integer indices to develop the index laws in algebraic form
- Establish that $x^0 = 1$ algebraically using index laws
- Simplify algebraic expressions that involve the zero index

Simplify algebraic products and quotients using index laws

- Simplify algebraic expressions that involve powers, products and quotients of simple algebraic terms containing positive-integer indices

Apply index laws to numerical expressions with negative-integer indices

- Apply index notation, patterns and index laws to establish the meaning of negative indices for numerical bases
- Evaluate numerical expressions involving a negative index by first representing them with a positive index
- Represent given numbers in index form (integer indices and bases only) and vice versa

Equations A

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- solves linear equations of up to 3 steps, limited to one algebraic fraction **MA5-EQU-C-01**

Related Life Skills outcomes: MALS-ADS-01, MALS-MDI-01

Content

Solve linear equations involving up to 3 steps

- Solve linear equations using algebraic techniques involving up to 3 steps
- Solve linear equations with pronumerals on both sides of the equation
- Solve linear equations involving grouping symbols
- Verify solutions using substitution

Solve linear equations involving one algebraic fraction

- Solve linear equations involving one algebraic fraction using up to 3 steps

Solve linear equations arising from word problems and substitution into formulas

- Solve linear equations arising from substitution into formulas
- Represent word problems as linear equations, solve the equations and interpret the solutions in the context of the problem

Linear relationships A

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- determines the midpoint, gradient and length of an interval, and graphs linear relationships, with and without digital tools **MA5-LIN-C-01**

Related Life Skills outcomes: MALS-POS-01

Content

Find the midpoint and gradient of a line segment (interval) on the Cartesian plane

- Plot and join 2 points to form an interval on the Cartesian plane and use the interval as the hypotenuse of a right-angled triangle
- Apply the relationship gradient $m = \frac{rise}{run}$ to find the gradient/slope of the interval joining the 2 points
- Distinguish between intervals with positive and negative gradients from a diagram
- Explain why horizontal intervals have a gradient of 0 and vertical intervals have undefined gradients using the gradient relationship
- Determine the midpoint of horizontal and vertical intervals on the Cartesian plane
- Apply the process for calculating the mean to find the midpoint, M of the interval joining 2 points on the Cartesian plane
- Use graphing applications to find the midpoint and gradient/slope of an interval

Find the distance between 2 points located on the Cartesian plane

- Use the interval between 2 points as the hypotenuse of a right-angled triangle on the Cartesian plane and apply Pythagoras' theorem to determine the length of the interval joining the 2 points
- Use graphing applications to find the distance between 2 points on the Cartesian plane

Recognise and graph equations

- Recognise that equations of the form $y = mx + c$ represent linear relationships or straight lines
- Construct tables of values and use coordinates to graph a variety of linear relationships on the Cartesian plane, with and without digital tools
- Identify the x- and y- intercepts of lines
- Determine whether a point lies on a line using substitution

Examine parallel, horizontal and vertical lines

- Explain that parallel lines have equal gradients/slopes
- Explain why the x-axis has the equation $y = 0$ and the y-axis has the equation $x = 0$
- Recognise $y = c$ as a line parallel to the x-axis and $x = k$ as a line parallel to the y-axis
- Graph vertical and horizontal lines

Linear relationships B

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- graphs and interprets linear relationships using the gradient/slope-intercept form **MA5-LIN-C-02**

Related Life Skills outcomes: MALS-POS-01

Content

Examine the gradient/slope-intercept form

- Interpret the coefficient of x(m) as the gradient/slope, and the constant (c) as the y-intercept for equations of the form $y = mx + c$
- Find the equation of a straight line in the form $y = mx + c$, given the gradient/slope and the y-intercept of the line
- Graph equations of the form $y = mx + c$ by using the gradient and the y-intercept
- Determine the gradient and y-intercept of a straight line from its graph and apply these values to determine the equation of the line
- Explain the effect of increasing or decreasing the gradient with or without digital tools
- Recognise and describe linear relationships in real-life contexts

Find the equations of parallel and perpendicular lines

- Explain that 2 straight lines are perpendicular if the product of their gradients is -1
- Find the equation of a straight line that is parallel or perpendicular to another given line by applying $y = mx + c$

Non-linear relationships A

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- identifies connections between algebraic and graphical representations of quadratic and exponential relationships in various contexts **MA5-NLI-C-01**

Content

Examine the connection between algebraic and graphical representations of quadratics and exponentials

- Represent quadratic and exponential relationships using graphing applications
- Construct a table of values to graph non-linear relationships involving quadratics and exponentials
- Explain that quadratic relationships are represented by parabolas
- Identify graphs and equations of parabolas and exponential curves
- Recognise quadratics and exponentials in real-life contexts

Non-linear relationships B

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- identifies and compares features of parabolas and exponential curves in various contexts **MA5-NLI-C-02**

Content

Graph and examine quadratic relationships

- Graph and compare parabolas of the form $y = kx^2$ and $y = kx^2 + c$ using graphing applications
- Identify and describe the key features of parabolas of the form $y = kx^2$ and $y = kx^2 + c$ including the vertex, x- and y-intercepts, axis of symmetry and concavity

Graph and examine exponential relationships

- Graph and compare exponential curves of the form $y = a^x$ using graphing applications
- Describe features of exponential curves including the y-intercept, asymptote and the nature of the curve for very large and very small values of x

Distinguish between linear, quadratic and exponential relationships by examining their graphical representations

- Associate graphs of straight lines, parabolas and exponential curves with the appropriate equations
- Recognise non-linear relationships in real-life contexts and solve related problems
- Use graphing applications to solve a pair of simultaneous equations where one equation is non-linear and interpret the solution

Numbers of any magnitude

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- solves measurement problems by using scientific notation to represent numbers and rounding to a given number of significant figures **MA5-MAG-C-01**

Content

Identify and describe very small and very large measurements

- Identify and describe the meaning of common prefixes, such as *milli*, *centi* and *kilo*
- Establish the meaning of prefixes for very small or very large measurement units

Find absolute and percentage error

- Determine the precision of a measuring instrument by finding the smallest division on the instrument
- Find the absolute error of measuring instruments ($absolute\ error = \frac{1}{2} \times precision$)
- Calculate the percentage error of a given measurement by applying the formula:

$$percentage\ error = \frac{absolute\ error}{measurement} \times 100\%$$

Estimate and round numbers to a specified degree of accuracy

- Apply the language of estimation appropriately, including the terms *rounding*, *approximate* and *level of accuracy*
- Round numbers to a specified number of significant figures
- Examine the effect that truncating or rounding during calculations has on the accuracy of the results

Express numbers in scientific notation

- Recognise the need for notation to express very large or very small numbers
- Represent numbers using scientific notation in practical contexts
- Order numbers expressed in scientific notation
- Represent numbers expressed in scientific notation as a decimal
- Estimate the value of calculations involving scientific notation by applying knowledge of index laws
- Solve problems with calculations involving scientific notation using digital tools

Trigonometry A

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- applies trigonometric ratios to solve right-angled triangle problems **MA5-TRG-C-01**

Content

Demonstrate and explain the constancy of trigonometric ratios for a given angle in right-angled triangles

- Identify and label the hypotenuse, adjacent and opposite sides with respect to a given angle in a right-angled triangle in any orientation
- Define the sine, cosine and tangent ratios for angles in right-angled triangles and use trigonometric notation $\sin\theta$, $\cos\theta$, $\tan\theta$
- Identify the sine, cosine and tangent ratios in a right-angled triangle
- Verify the constancy of the sine, cosine and tangent ratios for a given angle by applying knowledge of similar right-angled triangles
- Find approximations of the trigonometric ratios for a given angle
- Find the size of an angle given one of the trigonometric ratios for the angle using digital tools

Apply trigonometry to solve right-angled triangle problems

- Apply trigonometry to find the lengths of unknown sides in right-angled triangles with a given angle including angles in degrees and minutes
- Apply trigonometry to find the size of unknown angles in right-angled triangles including in degrees and minutes
- Solve a variety of practical problems involving trigonometric ratios in right-angled triangles

Trigonometry B

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- applies trigonometry to solve problems, including bearings and angles of elevation and depression **MA5-TRG-C-02**

Content

Solve right-angled triangle problems involving angles of elevation and depression

- Identify and describe angles of elevation and depression
- Solve practical problems involving angles of elevation and depression

Solve right-angled triangle problems involving bearings

- Identify and interpret true bearings and compass bearings
- Explain the difference between true bearings and compass bearings and convert between them
- Solve practical problems involving bearings

Area and surface area A

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- solves problems involving the surface area of right prisms and practical problems involving the area of composite shapes and solids **MA5-ARE-C-01**

Related Life Skills outcomes: MALS-ARE-01

Content

Solve problems involving areas and surface areas

- Solve practical problems involving the areas of composite shapes
- Identify the edge lengths and the faces making up the surface area of prisms
- Recognise and justify whether a diagram represents a net of a right prism
- Create and rearrange nets of right prisms
- Find the surface areas of prisms, given their nets, excluding curved surfaces
- Solve problems involving surface areas of prisms, excluding curved surfaces

Develop and apply the formula for surface areas of cylinders

- Recognise the curved surface of a cylinder as a rectangle and apply this knowledge to calculate the area of the curved surface
- Develop and apply the formula to find the surface area of a closed cylinder: $A = 2\pi r^2 + 2\pi rh$, where r is the length of the radius and h is the perpendicular height

Solve problems involving surface areas of cylinders and related composite solids

- Solve problems involving surface areas of cylinders and related composite solids

Volume A

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- solves problems involving the volume of composite solids consisting of right prisms and cylinders **MA5-VOL-C-01**

Related Life Skills outcomes: MALS-VOL-01

Content

Solve problems involving composite solids consisting of right prisms and cylinders

- Find the volumes of composite right prisms with uniform cross-sections that may be dissected into triangles and quadrilaterals
- Find the volumes of right prisms that have uniform cross-sections in the form of sectors, semicircles and quadrants
- Calculate volumes of composite solids consisting of right prisms and cylinders
- Solve practical problems related to the volumes and capacities of composite solids

Properties of geometrical figures A

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- identifies and applies the properties of similar figures and scale drawings to solve problems **MA5-GEO-C-01**

Related Life Skills outcomes: MALS-GEO-01

Content

Identify and describe the properties of similar figures

- Describe similar figures as having the same shape but not necessarily the same size
- Verify and explain that in similar polygons, the corresponding angles are equal and the corresponding side lengths are in the same proportion
- Name the vertices in matching order when using the similar symbol (~) in a similarity statement
- Match the corresponding sides and angles of similar polygons

Solve problems using ratio and scale factors in similar figures

- Apply an appropriate scale to enlarge or reduce a diagram
- Determine the scale factor for pairs of similar polygons and circles
- Apply knowledge of scale factor to find unknown sides in similar polygons
- Solve problems involving unknown lengths and scale factors of similar figures and related practical problems
- Solve problems involving scale drawings, with or without digital tools

Data analysis A

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- compares and analyses datasets using summary statistics and graphical representations **MA5-DAT-C-01**

Related Life Skills outcomes: MALS-DAT-02

Content

Examine standard deviation as a measure of spread

- Identify standard deviation as a measure of spread
- Calculate the standard deviation of a small dataset using digital tools
- Compare small datasets using standard deviation

Determine quartiles and interquartile range

- Determine the 5-number summary for sets of numerical data
- Determine the 5-number summary from graphical representations
- Determine the interquartile range (IQR) for datasets
- Compare and explain the relative merits of range and IQR as measures of spread

Represent datasets using box plots and use them to compare datasets

- Represent numerical datasets using a box plot to display the median, upper and lower quartiles, and maximum and minimum values
- Compare 2 or more numerical datasets using parallel box plots drawn on the same scale
- Compare and contrast the centres, spreads and shapes of 2 or more numerical datasets, using box plots and numerical statistics, including the 5-number summary
- Determine quartiles from datasets displayed in histograms and dot plots, and represent these as a box plot
- Identify and describe skewness or symmetry of datasets displayed in histograms, dot plots and box plots
- Interpret box plots to draw conclusions and make inferences about the dataset

Data analysis B

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- displays and interprets datasets involving bivariate data **MA5-DAT-C-02**

Related Life Skills outcomes: MALS-DAT-02

Content

Identify and describe numerical datasets involving 2 variables

- Distinguish between situations involving 1-variable and 2-variable (bivariate) data and explain when each is needed
- Explain the difference between variables that have an association and variables that have a causal relationship
- Identify and describe the independent variable and dependent variable in relationships with possible cause and effect

Represent datasets involving 2 numerical variables, using a scatter plot and a line of best fit, by eye

- Gather data on a topic of interest involving 2 numerical variables
- Represent the data using a scatter plot
- Create a line of best fit, by eye, on an existing scatter plot

Interpret data involving 2 numerical variables, using graphical representations

- Describe informally the association between 2 numerical variables and apply terminology about form (linear), strength (strong, moderate or weak) and direction (positive or negative)
- Use the line of best fit, by eye, to make predictions between known data values (interpolation) and what might happen beyond known data values (extrapolation)
- Explain the limitations of the model when making predictions

Probability A

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- solves problems involving probabilities in multistage chance experiments and simulations **MA5-PRO-C-01**

Related Life Skills outcomes: MALS-PRO-01

Content

Describe multistage chance experiments involving independent and dependent events

- Explain the difference between dependent and independent events in experiments involving 2 stages
- Explain how the probability of independent and dependent events differs in relation to replacement

Solve problems for multistage chance experiments

- Record all possible outcomes for multistage chance experiments
- Determine the probabilities of outcomes for multistage independent events using $P(A \text{ and } B) = P(A) \times P(B)$, where necessary
- Determine the probabilities of outcomes for multistage dependent events
- Associate complementary events with probabilities in multistage chance experiments

Design and use simulations to model and examine situations involving probability

- Design and conduct a probability simulation, modelling probabilities of events, using digital tools
- Record and use the results of a probability simulation to predict future events
- Apply reasoning to evaluate the simulation and its related outcomes

Variation and rates of change A (Path)

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- identifies and solves problems involving direct and inverse variation and their graphical representations *(Path: Stn, Adv)* **MA5-RAT-P-01**

Content

Stn (Standard), Adv (Advanced) and Ext (Extension) have been used to suggest paths for related Stage 6 courses.

Identify and describe problems involving direct and inverse variation

- Describe typical examples of direct variation/proportion
- Apply the language of direct variation to everyday contexts: y is directly proportional to x, y is proportional to x, y varies directly as x
- Identify and represent direct variation/proportion as $y \propto x$ (y is proportional to x) or $y = kx$, where k is the constant of variation
- Describe typical examples of inverse (indirect) variation
- Apply the language of inverse variation to everyday contexts: y is inversely proportional to x, y is proportional to the reciprocal of x, y varies inversely as x
- Identify and represent inverse variation/proportion as $y \propto \frac{1}{x}$ (y is inversely proportional to x) or $y = \frac{k}{x}$, where k is the constant of variation

Identify and describe graphs involving direct and inverse variation

- Recognise and describe direct variation from graphs, noting that the graph of $y = kx$ is a straight line passing through the origin, with its gradient k being the constant of variation
- Recognise and describe inverse variation from graphs, noting that the graph of $y = \frac{k}{x}$ is a curve

Solve problems involving direct and inverse variation and examine the relationship between graphs and equations corresponding to proportionality

- Solve problems involving direct or inverse variation using an equation
- Use linear conversion graphs to convert from one unit to another
- Graph equations representing direct variation, with or without digital tools

Variation and rates of change B (Path)

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- analyses and constructs graphs relating to rates of change *(Path: Stn, Adv)* **MA5-RAT-P-02**

Content

Stn (Standard), Adv (Advanced) and Ext (Extension) have been used to suggest paths for related Stage 6 courses.

Analyse graphs that are decreasing or increasing at a constant rate

- Describe direct variation graphs as increasing at a constant rate and interpret the gradient of the line as the rate of change
- Interpret and analyse further graphs that exhibit a constant rate of change, including those that are decreasing at a constant rate

Analyse the relationship between graphs and variable rates of change

- Interpret distance–time graphs when the speed is variable, describing the rate of increase or decrease, the initial and final points, constant relationships represented by straight lines and variable relationships represented by curved lines
- Describe the rate of change between variables in a variety of contexts including direct and inverse variation
- Describe qualitatively the rate of change of a graph, using terms such as *increasing at a decreasing rate*

Construct graphical representations of rates of change

- Represent a given description of a variable rate of change of one quantity over time

Algebraic techniques B (Path)

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- simplifies algebraic fractions involving indices, and expands and factorises algebraic expressions *(Path: Adv)* **MA5-ALG-P-01**

Content

Stn (Standard), Adv (Advanced) and Ext (Extension) have been used to suggest paths for related Stage 6 courses.

Apply the 4 operations involving algebraic fractions with pronumerals in the denominator

- Simplify algebraic fractions, including those involving indices
- Simplify expressions that involve operations with algebraic fractions, including algebraic fractions that involve pronumerals in the denominator and/or indices

Factorise algebraic expressions by taking out a common algebraic factor

- Factorise algebraic expressions, including those involving indices, by determining the highest common factors (HCF) including negative coefficients/pronumerals

Expand binomial products and factorise monic quadratic expressions

- Expand binomial products
- Factorise monic quadratic trinomial expressions
- Solve problems involving expansion and factorisation of algebraic expressions

Algebraic techniques C (Path)

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- selects and applies appropriate algebraic techniques to operate with algebraic fractions, and expands, factorises and simplifies algebraic expressions *(Path: Adv)* **MA5-ALG-P-02**

Content

Stn (Standard), Adv (Advanced) and Ext (Extension) have been used to suggest paths for related Stage 6 courses.

Operate with algebraic fractions involving binomial numerators and numerical denominators

- Add and subtract algebraic fractions with binomial numerators and numerical denominators

Expand, factorise and simplify algebraic expressions including special products

- Prove and apply these special products $(a-b)(a+b)=a^2-b^2$, $(a+b)^2=a^2+2ab+b^2$, $(a-b)^2=a^2-2ab+b^2$
- Expand and simplify a variety of algebraic expressions including binomial products and the special products
- Factorise algebraic expressions involving the special products and strategies, including common factors, grouping in pairs for 4-term expressions and quadratic trinomials (monic and non-monic)
- Simplify algebraic expressions involving algebraic fractions using factorisation

Indices B (Path)

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- applies the index laws to operate with algebraic expressions involving negative-integer indices *(Path: Adv)* **MA5-IND-P-01**

Content

Stn (Standard), Adv (Advanced) and Ext (Extension) have been used to suggest paths for related Stage 6 courses.

Apply index laws to algebraic expressions involving negative-integer indices

- Apply index notation, patterns and index laws to establish $a^{-1} = \frac{1}{a}$, $a^{-2} = \frac{1}{a^2}$, $a^{-3} = \frac{1}{a^3}$ and
$$a^{-n} = \frac{1}{a^n}$$
- Represent expressions involving negative-integer indices as expressions involving positive-integer indices and vice versa
- Apply the index laws to simplify algebraic products and quotients involving negative-integer indices
- Describe and use x^{-1} as the reciprocal of x and generalise this relationship to expressions of the form $\left(\frac{a}{b}\right)^{-1}$
- Use knowledge of the reciprocal to simplify expressions of the form $\left(\frac{a}{b}\right)^{-n}$

Indices C (Path)

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- describes and performs operations with surds and fractional indices *(Path: Adv)* **MA5-IND-P-02**

Content

Stn (Standard), Adv (Advanced) and Ext (Extension) have been used to suggest paths for related Stage 6 courses.

Describe surds

- Describe a real number as a number that can be represented by a point on the number line
- Examine the differences between rational and irrational numbers and recognise that all rational and irrational numbers are real
- Convert between recurring decimals and their fractional form using digital tools
- Describe the term *surd* as referring to irrational expressions of the form $\sqrt[n]{x}$ where x is a rational number and n is an integer such that $n \geq 2$, and $x > 0$ when n is even
- Recognise that a surd is an exact value that can be approximated by a rounded decimal
- Demonstrate that $\sqrt{x}$ is undefined for $x < 0$ and that $\sqrt{x} = 0$ when $x = 0$ using digital tools
- Describe $\sqrt{x}$ as the positive square root of x for $x > 0$ and $\sqrt{0} = 0$

Apply knowledge of surds to solve problems

- Establish and apply the following results for $x > 0$ and $y > 0$: $(\sqrt{x})^2 = x = \sqrt{x^2}$, $\sqrt{xy} = \sqrt{x} \times \sqrt{y}$ and $\sqrt{\frac{x}{y}} = \frac{\sqrt{x}}{\sqrt{y}}$
- Apply the 4 operations to simplify expressions involving surds
- Expand and simplify expressions involving surds
- Rationalise the denominators of surds of the form $\frac{a\sqrt{b}}{c\sqrt{d}}$

Describe and use fractional indices

- Apply index laws to describe fractional indices as: $a^{\frac{1}{n}} = \sqrt[n]{a}$ and $a^{\frac{m}{n}} = \sqrt[n]{a^m} = (\sqrt[n]{a})^m$
- Translate expressions in surd form to expressions in index form and vice versa
- Evaluate numerical expressions involving fractional indices including using digital tools

Equations B (Path)

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- solves monic quadratic equations, linear inequalities and cubic equations of the form $ax^3 = k$ *(Path: Adv)* **MA5-EQU-P-01**

Content

Stn (Standard), Adv (Advanced) and Ext (Extension) have been used to suggest paths for related Stage 6 courses.

Solve monic quadratic equations

- Solve quadratic equations of the form $ax^2 + bx + c = 0$, limited to $a = 1$, using factors

Solve cubic equations

- Determine that for any value of k there is a unique value of x that solves a cubic equation of the form $ax^3 = k$ where $a \neq 0$
- Solve cubic equations of the form $ax^3 = k$, leaving answers in exact form and as decimal approximations

Solve linear inequalities and graph their solutions on a number line

- Represent inequalities on a number line
- Solve linear inequalities, including those with negative numbers, and graph the solutions
- Recognise that an inequality has an infinite number of solutions unless other restrictions are introduced

Equations C (Path)

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- solves linear equations of more than 3 steps, monic and non-monic quadratic equations, and linear simultaneous equations *(Path: Adv)* **MA5-EQU-P-02**

Content

Stn (Standard), Adv (Advanced) and Ext (Extension) have been used to suggest paths for related Stage 6 courses.

Solve linear equations involving algebraic fractions and equations of more than 3 steps

- Solve linear equations involving more than 3 steps
- Solve equations that involve 2 or more fractions

Rearrange literal equations

- Change the subject of a formula

Solve quadratic equations using a variety of methods

- Solve equations of the form $ax^2 + bx + c = 0$ by factorisation and by completing the square
- Apply the quadratic formula $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ to solve quadratic equations
- Apply the most appropriate method to solve a variety of quadratic equations
- Use substitution to verify solutions to quadratic equations
- Identify whether a given quadratic equation has real solutions, and if there are real solutions, whether or not they are equal
- Solve quadratic equations resulting from substitution into formulas in various contexts
- Model and solve word problems using quadratic equations in various contexts
- Solve equations that are reducible to quadratics

Solve linear simultaneous equations, both algebraically and graphically

- Solve linear simultaneous equations by finding the point of intersection of their graphs
- Solve linear simultaneous equations using algebraic techniques including substitution and elimination methods
- Model and solve word problems using simultaneous equations and interpret their solutions
- Describe an identity as an equation that is true for all values of the pronumeral and relate the identity to coincident lines
- Describe a contradiction as an equation that has no solutions and relate the contradiction to parallel lines

Linear relationships C (Path)

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- describes and applies transformations, the midpoint, gradient/slope and distance formulas, and equations of lines to solve problems *(Path: Adv)* **MA5-LIN-P-01**

Content

Stn (Standard), Adv (Advanced) and Ext (Extension) have been used to suggest paths for related Stage 6 courses.

Apply formulas to find the midpoint and gradient/slope of an interval on the Cartesian plane

- Apply the formula to find the midpoint of the interval joining 2 points on the Cartesian plane: $M(x,y) = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$
- Use the relationship $m = \frac{\text{rise}}{\text{run}}$ to establish the formula for the gradient/slope (m) of the interval joining the 2 points (x_1, y_1) and (x_2,y_2) on the Cartesian plane: $m = \frac{y_2 - y_1}{x_2 - x_1}$
- Apply the gradient formula to find the gradient of the interval joining 2 points on the Cartesian plane

Apply the distance formula to find the distance between 2 points located on the Cartesian plane

- Apply knowledge of Pythagoras' theorem to establish the formula for the distance (d) between the 2 points (x_1, y_1) and (x_2,y_2) on the Cartesian plane: $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
- Apply the distance formula to find the distance between 2 points on the Cartesian plane

Use various forms of the equation of a straight line

- Rearrange linear equations from gradient–intercept form $(y = mx + c)$ to general form $(ax + by + c = 0)$ and vice versa
- Find the x- and y-intercepts of a straight line in any form
- Graph the equation of a straight line in any form
- Use the point–gradient form $(y - y_1 = m(x - x_1))$ or the gradient–intercept form $(y = mx + c)$ to find the equation of a line passing through a point (x_1, y_1), with a given gradient m
- Use the gradient and the point–gradient form to find the equation of a line passing through 2 points
- Find the equation of a line that is parallel or perpendicular to a given line in any form
- Determine and justify whether 2 given lines are parallel or perpendicular

Solve problems by applying coordinate geometry formulas

- Solve problems including those involving geometrical figures by applying coordinate geometry formulas

Identify line and rotational symmetries

- Identify lines (axes) and rotational symmetry in plane shapes

- Identify line and rotational symmetry in various linear and non-linear graphs

Describe translations, reflections in an axis, and rotations through multiples of 90 degrees on the Cartesian plane, using coordinates

- Apply the notation P' to name the image resulting from applying a transformation to a point P on the Cartesian plane
- Determine and plot the coordinates for P' resulting from translating P one or more times
- Determine and plot the coordinates for P' resulting from reflecting P in either the x- or y-axis
- Determine and plot the coordinates for P' resulting from rotating P by a multiple of 90° about the origin

Non-linear relationships C (Path)

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- interprets and compares non-linear relationships and their transformations, both algebraically and graphically *(Path: Adv)* **MA5-NLI-P-01**

Content

Stn (Standard), Adv (Advanced) and Ext (Extension) have been used to suggest paths for related Stage 6 courses.

Graph parabolas and describe their features and transformations

- Use graphing applications to compare parabolas of the form $y = kx^2$, $y = kx^2 + c$, $y = k(x - b)^2$ and $y = k(x - b)^2 + c$, and describe their features and transformations
- Find x- and y-intercepts algebraically, where appropriate, for the graph of $y = ax^2 + bx + c$, given a,b and c
- Determine the equation of the axis of symmetry of a parabola using either the formula $x = -\frac{b}{2a}$ or the midpoint of the x-intercepts
- Find the coordinates of a parabola's vertex using a variety of methods
- Graph quadratic relationships of the form $y = ax^2 + bx + c$ by identifying and applying features of parabolas and their equations without graphing software

Graph exponentials and describe their features and transformations

- Use graphing applications to graph exponential relationships of the form $y = k(a)^x + c$ and $y = k(a)^{-x} + c$ for integer values of k, a and c (where $a > 0$ and $a \neq 1$), and compare and describe any relevant features

Graph hyperbolas and describe their features and transformations

- Use graphing applications to graph, compare and describe hyperbolic relationships of the form $y = \frac{k}{x}$ for integer values of k
- Use graphing applications to graph and describe a variety of hyperbolas, including where the equation is given in the form $y = \frac{k}{x} + c$ or $y = \frac{k}{x - b}$ for integer values of k, b and c

Graph circles and describe their features and transformations

- Derive the equation of a circle $x^2 + y^2 = r^2$ with centre $(0, 0)$ and radius r using the distance formula
- Identify and describe equations that represent circles with centre at the origin and radius of the circle r
- Graph circles of the form $x^2 + y^2 = r^2$, where r is the radius of the circle using graphing applications
- Establish the equation of the circle with centre (a,b) and radius r, and graph equations of the form $(x - a)^2 + (y - b)^2 = r^2$

- Find the centre and radius of a circle with the equation in the form $x^2 + y^2 + ax + by + c = 0$ by completing the square

Distinguish between different types of graphs by examining their algebraic and graphical representations and solve problems

- Identify and describe features of different types of graphs based on their equations
- Identify a possible equation from a graph and verify using graphing applications
- Find points where a line intersects with a parabola, hyperbola or circle, both graphically and algebraically

Graph and compare polynomial curves and describe their features and transformations

- Use graphing applications to graph and compare features of cubic equations of the form $y = ax^3 + c$, where a and c are integers
- Use graphing applications to graph a variety of equations of the form $y = kx^n$, where n is an integer and $n \geq 2$, and describe the effect on the shape of the curve where n is an odd or an even number
- Use graphing applications to graph curves of the form $y = kx^n + c$ and $y = k(x - b)^n$ where n is an integer and $n \geq 2$, and describe the transformations from $y = kx^n$

Polynomials (Path)

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- defines, operates with and graphs polynomials and applies the factor and remainder theorems to solve problems *(Path: Adv, Ext)* **MA5-POL-P-01**

Content

Stn (Standard), Adv (Advanced) and Ext (Extension) have been used to suggest paths for related Stage 6 courses.

Define and operate with polynomials

- Recognise a polynomial expression $a_nx^n + a_{n-1}x^{n-1} + \ldots + a_2x^2 + a_1x + a_0$ where $n = 0, 1, 2\ldots$ and $a_0, a_1, a_2, \ldots, a_n$ are real numbers
- Describe polynomials using terms such as degree, leading term, coefficient and leading coefficient, constant term, monic and non-monic
- Define a monic polynomial as having a leading coefficient of one
- Apply the notation $P(x)$ for polynomials and $P(c)$ to indicate the value of $P(x)$ for $x = c$
- Add, subtract and multiply polynomials

Divide polynomials

- Identify the dividend, divisor, quotient and remainder in numerical division
- Divide a polynomial by a linear polynomial to find the quotient and remainder
- Express a polynomial in the form $P(x) = D(x)Q(x) + R(x)$, where $D(x)$ is the divisor, $Q(x)$ is the quotient and $R(x)$ is the remainder

Apply the factor and remainder theorems to solve problems

- Verify the remainder theorem and use it to find factors of polynomials and solve related problems
- Develop and apply the factor theorem to factorise particular polynomials completely and solve related problems
- Apply the factor theorem and division to find the zeroes of a polynomial $P(x)$ and solve $P(x) = 0$ (degree ≤ 4)
- State the maximum number of zeroes a polynomial of degree n can have

Graph polynomials

- Graph polynomials in factored form
- Graph quadratic, cubic and quartic polynomials by factorising and finding the zeroes
- Relate the term *zeroes* to polynomial functions and *roots* to polynomial equations
- Use graphing applications to determine the effect of single, double and triple roots of a polynomial equation $P(x) = 0$ on the shape of the graph for $y = P(x)$
- Graph polynomials using the sign of the leading term and the multiplicity of roots for the equation $P(x) = 0$
- Use graphing applications to compare the graphs of $y = -P(x)$, $y = P(-x)$, $y = P(x) + c$ and $y = kP(x)$ to the graph of $y = P(x)$

Logarithms (Path)

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- establishes and applies the laws of logarithms to solve problems *(Path: Adv)* **MA5-LOG-P-01**

Content

Stn (Standard), Adv (Advanced) and Ext (Extension) have been used to suggest paths for related Stage 6 courses.

Examine logarithms both numerically and graphically

- Define the term *logarithm*: the logarithm of a number to any positive base a is the index to which a is raised to give this number
- Recognise equivalence where $y = a^x$ is equivalent to $x = \log_a y$ where $a > 0$ and $a \neq 1$
- Translate statements expressing a number in index form into equivalent statements expressing the logarithm of the number
- Use graphing applications to compare and contrast graphs for the functions $y = a^x$ and $y = \log_a x$
- Generalise that $y = \log_a x$ is an increasing function when $a > 1$ and decreasing when $0 < a < 1$

Establish and apply the laws of logarithms to solve problems

- Deduce laws of logarithms from laws of indices
- Establish and use a variety of logarithmic results
- Apply the laws of logarithms to evaluate and simplify expressions
- Solve simple equations that involve exponents or logarithms
- Examine logarithmic scales and explain their use in various contexts

Functions and other graphs (Path)

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- uses function notation to describe and graph functions of one variable and graphs inequalities in one and 2 variables *(Path: Adv)* **MA5-FNC-P-01**

Content

Stn (Standard), Adv (Advanced) and Ext (Extension) have been used to suggest paths for related Stage 6 courses.

Define relations and functions, and use function notation

- Define a relation over 2 sets as an association between the elements of one set to the elements of another set
- Notate relations as a set of ordered pairs (x,y) of real numbers
- Define a function as a relation over 2 sets where each element of the first set is associated with exactly one element in the second set
- Apply the vertical line test on a graph to decide whether it represents a function
- Use the notation $f(x)$ when expressing a function
- Use the notation $f(c)$ to determine the value of $f(x)$ when $x = c$

Find the domain and range of a function and graph functions

- Define the domain as the set of all allowable values of x
- Define the range as the set of possible y-values as x varies over the domain of the function
- Determine and describe the domain and range for a variety of functions
- Use graphing applications to graph and compare functions of the form $y = f(x)$, $y = f(x) + c$, $y = f(x - b)$, $y = kf(x)$ and $y = f(ax)$, and describe their transformations

Graph regions corresponding to linear inequalities in one and 2 variables

- Graph linear inequalities of the form $ax + by > c$, testing whether the points satisfy the given inequality and shading appropriate regions

Trigonometry C (Path)

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- applies Pythagoras' theorem and trigonometry to solve 3-dimensional problems and applies the sine, cosine and area rules to solve 2-dimensional problems, including bearings *(Path: Stn, Adv)* **MA5-TRG-P-01**

Content

Stn (Standard), Adv (Advanced) and Ext (Extension) have been used to suggest paths for related Stage 6 courses.

Solve 3-dimensional problems involving right-angled triangles

- Apply Pythagoras' theorem to solve problems involving the lengths of the edges and diagonals of rectangular prisms and other 3-dimensional objects
- Apply trigonometry to solve problems involving right-angled triangles in 3 dimensions, including using bearings and angles of elevation and depression

Apply the sine, cosine and area rules to any triangle and solve related problems

- Use graphing applications to verify the sine rule $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$ and that the ratios of a side to the sine of the opposite angle is a constant
- Apply the sine rule in a given triangle ABC to find the value of an unknown side
- Apply the sine rule in a given triangle ABC to find the value of an unknown angle (ambiguous case excluded): $\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$
- Use graphing applications to verify the cosine rule $c^2 = a^2 + b^2 - 2ab\cos C$.
- Apply the cosine rule to find the unknown sides for a given triangle ABC.
- Rearrange the formula to deduce that $\cos C = \frac{a^2 + b^2 - c^2}{2ab}$ and use this to find an unknown angle
- Use graphing applications to verify the area rule $A = \frac{1}{2}ab\sin C$.
- Apply the formula $A = \frac{1}{2}ab\sin C$, where a and b are the sides that form angle C to find the area of a given triangle ABC
- Solve problems involving finding unknown angles or sides in triangles (excluding right-angled triangles) by selecting and applying the appropriate rule

Trigonometry D (Path)

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- establishes and applies the properties of trigonometric functions and finds solutions to trigonometric equations *(Path: Adv)* **MA5-TRG-P-02**

Content

Stn (Standard), Adv (Advanced) and Ext (Extension) have been used to suggest paths for related Stage 6 courses.

Use the unit circle to define trigonometric functions and represent them graphically

- Redefine the sine and cosine ratios in terms of the unit circle
- Verify that the tangent ratio can be expressed as a ratio of the sine and cosine ratios
- Use graphing applications to examine the sine, cosine and tangent ratios for (at least) $0^\circ \leq x \leq 360^\circ$, and graph the results
- Use graphing applications to examine graphs of the sine, cosine and tangent functions for angles of any magnitude, including negative angles
- Use the unit circle or graphs of trigonometric functions to establish and apply the relationships $\sin A = \sin(180° - A)$, $\cos A = -\cos(180° - A)$, and $\tan A = -\tan(180° - A)$ for obtuse angles when $0° \leq A \leq 90°$
- Establish and apply the relationship $m = \tan\theta$, where m is the gradient of the line and θ is the angle of inclination of a line with the x-axis on the Cartesian plane

Solve trigonometric equations using exact values and the relationships between supplementary and complementary angles

- Derive and apply the exact sine, cosine and tangent ratios for angles of $30°$, $45°$ and $60°$
- Verify and use the relationships between the sine and cosine ratios of complementary angles in right-angled triangles: $\sin A = \cos(90° - A)$ and $\cos A = \sin(90° - A)$
- Find the possible acute and/or obtuse angles, given a trigonometric ratio
- Apply the sine rule and area rule to find angles involving the ambiguous case

Area and surface area B (Path)

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- applies knowledge of the surface area of right pyramids and cones, spheres and composite solids to solve problems *(Path: Stn, Adv)* **MA5-ARE-P-01**

Related Life Skills outcomes: MALS-ARE-01

Content

Stn (Standard), Adv (Advanced) and Ext (Extension) have been used to suggest paths for related Stage 6 courses.

Solve problems involving surface areas

- Identify the perpendicular heights and slant heights of right pyramids and right cones
- Apply Pythagoras' theorem to find the slant heights, base lengths and perpendicular heights of right pyramids and right cones
- Solve problems involving the surface areas of right pyramids and compare methods of solution
- Apply the formula to find the curved surface area of right cones: $A = \pi rl$, where r is the length of the radius and l is the slant height
- Apply the formula to find the surface area of spheres: $A = 4\pi r^2$, where r is the length of the radius
- Solve problems involving the surface area of solids in a variety of contexts including composite solids

Volume B (Path)

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- applies knowledge of the volume of right pyramids, cones and spheres to solve problems involving related composite solids *(Path: Stn, Adv)* **MA5-VOL-P-01**

Content

Stn (Standard), Adv (Advanced) and Ext (Extension) have been used to suggest paths for related Stage 6 courses.

Solve problems involving volumes

- Find the volume of right pyramids and right cones by using an appropriate formula: volume of pyramid or cone $V = \frac{1}{3}Ah$, where A is the base area and h is the perpendicular height
- Find the volume of spheres by using an appropriate formula: volume of sphere $V = \frac{4}{3}\pi r^3$, where r is the length of the radius
- Apply knowledge of right pyramids, right cones and hemispheres to solve problems involving composite solids
- Solve practical problems related to the volumes and capacities of solids including right pyramids, right cones and spheres

Properties of geometrical figures B (Path)

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- establishes conditions for congruent triangles and similar triangles and solves problems relating to properties of similar figures and plane shapes *(Path: Ext)* **MA5-GEO-P-01**

Content

Stn (Standard), Adv (Advanced) and Ext (Extension) have been used to suggest paths for related Stage 6 courses.

Identify and explain congruence

- Identify figures as congruent figures if translations, reflections and rotations can move one figure exactly on top of another
- Match the sides and angles of 2 congruent polygons
- Indicate 2 polygons are congruent using the symbol $(\equiv)$ and name the vertices in matching order
- Determine that having equal radii is the condition for 2 circles to be congruent

Develop and use the conditions for congruent triangles

- Establish the 4 congruence tests for triangles (SSS, SAS, AAS and RHS)
- Use the congruence tests to identify a pair of congruent triangles from a selection of 2 or more triangles

Develop and apply the minimum conditions for triangles to be similar

- Examine the minimum conditions needed and establish the 4 tests for 2 triangles to be similar
- Apply the minimum conditions needed and determine whether 2 triangles are similar using an appropriate test

Establish and apply properties of similar shapes and solids

- Establish for 2 similar figures with similarity ratio $a:b$ that their areas are in the ratio $a^2:b^2$ and their volumes are in the ratio $a^3:b^3$
- Solve problems involving areas and volumes of similar shapes and solids

Apply logical reasoning to numerical problems involving plane shapes

- Apply geometrical facts, properties and relationships to find the sizes of unknown sides and angles of plane shapes in diagrams, providing appropriate reasons
- Define the exterior angle of a convex polygon
- Establish the sum of the exterior angles of any convex polygon is $360°$ and verify this result
- Apply the result for the sum of the interior angles of a triangle to find, by dissection, the sum of the interior angles of a polygon with more than 3 sides
- Apply the results for the interior and exterior angles of polygons to solve problems involving polygons

Properties of geometrical figures C (Path)

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- constructs proofs involving congruent triangles and similar triangles and proves properties of plane shapes *(Path: Ext)* **MA5-GEO-P-02**

Content

Stn (Standard), Adv (Advanced) and Ext (Extension) have been used to suggest paths for related Stage 6 courses.

Construct formal proofs involving congruent and similar triangles

- Construct formal proofs of the congruence of triangles, preserving the matching order of vertices and drawing relevant conclusions
- Construct formal proofs of the similarity of triangles, preserving the matching order of vertices and drawing relevant conclusions

Apply logical reasoning to proofs involving plane shapes

- Recognise that a definition is the minimum amount of information needed to identify a particular figure
- Prove the properties of isosceles and equilateral triangles and special quadrilaterals from the formal definition of the shapes
- Prove and apply theorems and properties related to triangles and quadrilaterals
- Prove and apply tests for quadrilaterals
- Solve numerical and non-numerical problems in Euclidean geometry based on known assumptions and proven theorems

Circle geometry (Path)

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- applies deductive reasoning to prove circle theorems and solve related problems *(Path: Ext)* **MA5-CIR-P-01**

Content

Stn (Standard), Adv (Advanced) and Ext (Extension) have been used to suggest paths for related Stage 6 courses.

Prove and apply angle and chord properties of circles

- Apply terminology associated with angles in circles
- Identify the arc on which an angle at the centre or circumference stands
- Demonstrate that at any point on a circle there is a unique tangent to the circle, and that this tangent is perpendicular to the radius at the point of contact
- Prove chord properties of circles
- Prove angle properties of circles
- Apply chord and angle properties of circles to find unknown angles and lengths in diagrams

Prove and apply tangent and secant properties of circles

- Prove tangent and secant properties of circles
- Apply tangent and secant properties of circles to find unknown angles and lengths in diagrams

Introduction to networks (Path)

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- solves problems involving the characteristics of graphs/networks, planar graphs and Eulerian trails and circuits *(Path: Stn)* **MA5-NET-P-01**

Content

Stn (Standard), Adv (Advanced) and Ext (Extension) have been used to suggest paths for related Stage 6 courses.

Examine and describe a graph/network

- Describe a network as a collection of objects (nodes or vertices) interconnected by lines (edges) that can represent systems in the real world
- Examine real-world applications of networks such as social networks, supply chain networks and communication infrastructure, and explore other applications of networks
- Explain that the terms *graph* and *network* are interchangeable
- Identify and define elements of a graph including vertex, edge and degree
- Explain that a given graph can be drawn in different ways

Define a planar graph and apply Euler's formula for planar graphs

- Define a planar graph as any graph that can be drawn in the plane so that no 2 edges cross
- Define a non-planar graph as a graph that can never be drawn in the plane without some edges crossing
- Demonstrate that some graphs that have crossing edges are still planar if they can be redrawn so that no 2 edges cross
- Identify the number of faces in a planar graph
- Describe and apply Euler's formula for planar graphs: $v - e + f = 2$, where v = vertices, e = edges and f = faces

Explain the concept of Eulerian trails and circuits in the context of the Königsberg bridges problem

- Explain that a connected graph is a graph that is in one piece, so that any 2 vertices are connected by a path
- Define a walk on a graph to be a sequence of vertices and edges of a graph
- Explain the difference between trail, circuit, path and cycle
- Relate the definition of a trail to an Eulerian trail as a walk in which every edge in the graph is included exactly once
- Relate the definition of a circuit to an Eulerian circuit that is defined as an Eulerian trail that ends at its starting point
- Relate Euler's Seven Bridges of Königsberg network problem to the definition of an Eulerian trail or circuit

Data analysis C (Path)

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- plans, conducts and reviews a statistical inquiry into a question of interest *(Path: Stn, Adv)* **MA5-DAT-P-01**

Content

Stn (Standard), Adv (Advanced) and Ext (Extension) have been used to suggest paths for related Stage 6 courses.

Plan and conduct a statistical inquiry into a question of interest

- Design an aim and hypothesis based on a question of interest
- Apply ethical and efficient methods for gathering and organising data
- Produce a report containing multiple data visualisations that provide insights and information

Examine reports of studies in digital media and elsewhere for information on their planning and implementation

- Examine and evaluate the appropriateness of sampling methods and sample size in reports with statements about a population and how they can affect the results of a survey
- Critically review surveys, polls and media reports for accuracy and/or bias
- Examine the use of statistics and the associated probabilities in shaping decisions made by governments and companies

Probability B (Path)

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- solves problems involving Venn diagrams, 2-way tables and conditional probability *(Path: Adv)* **MA5-PRO-P-01**

Content

Stn (Standard), Adv (Advanced) and Ext (Extension) have been used to suggest paths for related Stage 6 courses.

Solve problems involving Venn diagrams and 2-way tables

- Represent and interpret data in Venn diagrams for mutually exclusive and non-mutually exclusive events
- Construct Venn diagrams to represent all possible combinations of 2 attributes from given or collected data
- Interpret data in 2-way tables to represent relationships between attributes
- Construct 2-way tables to represent the relationships between attributes
- Convert between representations of the relationships between 2 attributes in Venn diagrams and 2-way tables
- Define a set as a collection of distinct objects
- Use Venn diagrams, set language and notation for events, including $\bar{A}$ (or A' or A^c) for the complement of an event A, $A \cap B$ for 'A and B' (the intersection of events A and B) and $A \cup B$ for 'A or B' (the union of events A and B) and recognise mutually exclusive events

Use the language, 'if … then', 'given', 'of' and 'knowing that', to examine conditional statements and identify common mistakes in interpreting the language

- Calculate the probabilities of events where a condition restricts the sample space
- Describe the effect of a given condition on the sample space
- Identify conditional statements used in descriptions of chance situations
- Explain the validity of conditional statements when describing chance situations, referring to dependent and independent events
- Identify and explain common misconceptions relating to chance experiments

Describe mutually and non-mutually exclusive events using specific language and calculate related probabilities

- Explain the difference between mutually exclusive and non-mutually exclusive events
- Describe compound events using the terms *inclusive or* and *exclusive or*
- Describe non-mutually exclusive events using the terminology *and*, *inclusive or* and *exclusive or*
- Describe events using the terms *at least*, *at most*, *not* and *and*
- Calculate the probability of compound events using strategies including Venn diagrams and 2-way tables

Life Skills for Stages 4/5

Life Skills table of outcomes

MAO-WM-01 Working mathematically

develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly

This outcome is aligned to all content in each Stage.

Focus area	Life Skills for Stages 4/5
Language of number	**MALS-LAN-01** recognises language that represents number **MALS-LAN-02** responds to and uses language that represents number
Counting	**MALS-COU-01** counts in everyday contexts
Representing number	**MALS-REP-01** represents number in everyday contexts
Comparing and ordering	**MALS-COM-01** compares and orders numbers
Fractions	**MALS-FRC-01** demonstrates knowledge of fractions in everyday contexts
Decimals and percentages	**MALS-DEP-01** demonstrates knowledge of decimals and percentages in everyday contexts
Addition and subtraction	**MALS-ADS-01** uses strategies for addition and subtraction
Multiplication and division	**MALS-MDI-01** uses strategies for multiplication and division
Recognising money	**MALS-FIN-01** demonstrates knowledge of money in everyday contexts
Financial management	**MALS-FIN-02** plans and manages personal finances
Patterns	**MALS-PAT-01** recognises and applies patterns in everyday contexts
Recognising and interpreting time Includes: Recognising time Interpreting time	**MALS-TIM-01** demonstrates knowledge of time in everyday contexts **MALS-TIM-02** organises and measures time in everyday contexts

Focus area	Life Skills for Stages 4/5
Length	**MALS-LEN-01** measures and uses length in everyday contexts
Volume, capacity and mass	**MALS-VOL-01** measures and uses volume, capacity and mass in everyday contexts
Area	**MALS-ARE-01** measures and uses area in everyday contexts
Geometry	**MALS-GEO-01** explores 2-dimensional shapes and 3-dimensional objects
Position	**MALS-POS-01** demonstrates knowledge of position and direction in everyday contexts
Representing data	**MALS-DAT-01** recognises and represents data in everyday contexts
Interpreting data	**MALS-DAT-02** interprets information from data displays
Chance and probability	**MALS-PRO-01** applies chance and probability to everyday events

Life Skills outcomes and content for Stage 4/5

Language of number

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- recognises language that represents number **MALS-LAN-01**
- responds to and uses language that represents number **MALS-LAN-02**

Related Stage 4/5 outcomes: MA4-INT-C-01

Content

Respond to language related to number

- Recognise language representing quantities
- Respond to language used to describe quantities
- Compare amounts using language that describes number
- Recognise how languages other than English, including Aboriginal and Torres Strait Islander Languages, use language related to number

Recognise number words

- Recognise numbers
- Respond to directions involving number words
- Use number words to communicate meaning

Recognise ordinal terms

- Respond to directions using ordinal terms
- Use ordinal terms in everyday contexts

Counting

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- counts in everyday contexts **MALS-COU-01**

Related Stage 4/5 outcomes: MA4-INT-C-01

Content

Use a variety of counting strategies

- Count forwards and backwards from a given number
- Count with 1:1 correspondence
- Apply cardinality to demonstrate that the last number spoken in a counting sequence names the quantity for that set
- Apply counting skills to single-digit, 2-digit and 3-digit numbers
- Count by twos, fives, tens and hundreds
- Count forwards and backwards from zero
- Solve problems involving counting

Representing number

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- represents number in everyday contexts **MALS-REP-01**

Related Stage 4/5 outcomes: MA4-INT-C-01

Content

Recognise numbers in everyday contexts

- Recognise the numbers 0 to 9
- Recognise 2-digit numbers
- Recognise odd and even numbers
- Identify ways numbers are used in everyday life
- Instantly recognise the number of items in a group without counting (subitise)

Apply numbers in everyday contexts

- Record numbers
- Use numbers in everyday contexts
- Identify a range of number configurations

Comparing and ordering

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- compares and orders numbers **MALS-COM-01**

Related Stage 4/5 outcomes: MA4-INT-C-01

Content

Estimate and compare amounts

- Compare the number of items in a group to solve problems
- Estimate and compare quantities of items

Order according to numerical value

- Order groups of objects in ascending or descending order
- Order numbers
- Use ordinal terms
- Identify the value of a given number in relation to other numbers
- Demonstrate understanding of values greater than, equal to and less than zero

Fractions

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- demonstrates knowledge of fractions in everyday contexts **MALS-FRC-01**

Related Stage 4/5 outcomes: MA4-FRC-C-01

Content

Identify the fraction 'a half' in everyday contexts

- Recognise halves
- Apply the terms *half* and *halves*

Identify the fraction 'a quarter' in everyday contexts

- Recognise quarters
- Apply the term *quarter*

Identify fractions with different denominators in everyday contexts

- Recognise unit fractions
- Recognise non-unit fractions where the numerator is a number other than one

Use fractions in everyday contexts

- Recognise fractions in everyday contexts
- Compare fractions
- Calculate fractions, with or without technology
- Apply fractions to solve problems

Decimals and percentages

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- demonstrates knowledge of decimals and percentages in everyday contexts **MALS-DEP-01**

Related Stage 4/5 outcomes: MA4-FRC-C-01

Content

Identify percentages

- Identify percentages in everyday contexts
- Represent the size of percentages
- Match percentages to common fractions

Use percentages

- Estimate using percentages in everyday contexts
- Calculate percentages, with or without technology
- Apply percentages to solve problems

Identify decimals

- Recognise decimal numbers in everyday contexts
- Matches decimals to common fractions

Use decimals

- Estimate, using decimals, in everyday contexts
- Calculate decimals, with or without technology

Addition and subtraction

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- uses strategies for addition and subtraction **MALS-ADS-01**

Related Stage 4/5 outcomes: MA4-INT-C-01, MA4-RAT-C-01, MA4-EQU-C-01, MA5-EQU-C-01

Content

Identify the language used in addition and subtraction

- Apply the language of addition and subtraction in everyday contexts
- Recognise and use the symbols +, − and =

Apply strategies for addition

- Model addition to solve problems
- Apply basic facts of addition, sums with 2 single digits, and their corresponding subtraction fact
- Add 2 numbers
- Estimate and check the sum of 2 numbers
- Calculate missing numbers in a number sentence
- Select and apply appropriate strategies for addition to solve problems in a range of contexts

Apply strategies for subtraction

- Model subtraction to solve problems
- Subtract a number from another given number
- Estimate and check for subtraction
- Calculate missing numbers in a number sentence
- Select and apply appropriate strategies for subtraction to solve problems

Multiplication and division

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- uses strategies for multiplication and division **MALS-MDI-01**

Related Stage 4/5 outcomes: MA4-INT-C-01, MA4-RAT-C-01, MA4-EQU-C-01, MA5-EQU-C-01

Content

Identify the language used in multiplication and division

- Respond to and use the language of multiplication and division in everyday contexts
- Recognise and use the symbols $\times$, $\div$ and $=$

Apply strategies for multiplication

- Model multiplication by recognising equal groups or multiples of items to solve problems
- Multiply 2 numbers
- Estimate and check the product of 2 numbers
- Use single-digit multiplication to solve problems
- Use commutative properties of numbers
- Select and apply appropriate strategies for multiplication to solve problems

Apply strategies for division

- Model division by recognising equal groups to solve problems
- Divide 2 numbers
- Estimate the result when dividing 2 numbers
- Use multiplication and division as inverse operations
- Select and apply appropriate strategies for division to solve problems

Recognising money

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- demonstrates knowledge of money in everyday contexts **MALS-FIN-01**

Related Stage 4/5 outcomes: MA5-FIN-C-01, MA5-FIN-C-02

Content

Recognise when money is used in everyday contexts

- Recognise when payment is needed for purchases
- Recognise symbols and signs in the community associated with money
- Respond to language used to describe money and purchases

Recognise different amounts of money

- Identify Australian coins and notes
- Recognise that different currencies are used around the world
- Identify amounts in dollars and cents
- Record amounts in dollars and cents

Compare and order money

- Recognise that coins and notes have different values
- Order coins and notes

Financial management

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- plans and manages personal finances **MALS-FIN-02**

Related Stage 4/5 outcomes: MA5-FIN-C-01, MA5-FIN-C-02

Content

Purchase goods or services

- Use an appropriate payment method
- Recognise language associated with financial management
- Make informed decisions about purchases
- Demonstrate knowledge about the cost of goods and/or services
- Recognise the value of money when making purchases

Estimate with money

- Estimate amounts of money needed to purchase goods or services
- Estimate the amount of change due

Identify personal finances

- Recognise sources of personal finance
- Identify essential personal purchases

Demonstrate understanding of managing personal finance

- Keep money in a secure location
- Track income and spending
- Calculate expenses with available funds
- Respond to financial terms
- Interpret information from a variety of bills or receipts

Patterns

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- recognises and applies patterns in everyday contexts **MALS-PAT-01**

Related Stage 4/5 outcomes: MA4-ALG-C-01, MA5-ALG-C-01

Content

Recognise patterns

- Recognise patterns in the environment
- Match patterns, with or without technology

Describe and continue patterns

- Describe a pattern in a variety of contexts
- Continue patterns, with or without technology
- Identify an inconsistency in a pattern
- Describe and continue number patterns that increase or decrease

Copy and create a pattern

- Copy a pattern
- Create number patterns
- Apply knowledge of patterns

Recognising time

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- demonstrates knowledge of time in everyday contexts **MALS-TIM-01**

Content

Recognise the language of time

- Recognise that activities occur at different times of the day
- Follow instructions to participate in activities
- Recognise the language used to describe time

Use language of time

- Apply informal language of time to different times of day
- Identify the order of activities
- Describe and compare the duration of activities

Interpreting time

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- organises and measures time in everyday contexts **MALS-TIM-02**

Content

Relate times to personal events

- Identify when an activity will occur
- Recognise what time of day an activity will occur
- Respond to questions about when familiar activities occur

Organise personal events

- Create a personal schedule
- Identify days, weeks and months to organise personal events
- Associate activities with days, weeks and months
- Communicate dates for events of personal significance

Manage personal time

- Identify time associated with personal events
- Plan personal time for a range of activities
- Use knowledge of time to inform decisions

Interpret time

- Recognise and deal with time in a range of formats
- Relate *hour*, *half hour*, *quarter to* and *quarter past* to a variety of contexts
- Identify *am* and *pm*
- Recognise time in a range of formats
- Describe the duration of events

Estimate and measure time

- Plan a journey, with or without technology
- Select appropriate units of time for an activity
- Estimate time using a range of formats
- Estimate the duration of an activity
- Calculate the duration of an activity

Length

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- measures and uses length in everyday contexts **MALS-LEN-01**

Related Stage 4/5 outcomes: MA4-LEN-C-01

Content

Recognise features associated with length

- Identify features associated with length
- Recognise features associated with length that can be estimated and measured

Estimate and measure using informal and formal units

- Use informal units to estimate and measure length
- Recognise appropriate formal units for measuring length, including their abbreviations
- Measure and compare length using formal units

Estimate and measure length to solve problems

- Compare the length of different objects to solve problems
- Apply length to solve problems

Volume, capacity and mass

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- measures and uses volume, capacity and mass in everyday contexts **MALS-VOL-01**

Related Stage 4/5 outcomes: MA4-VOL-C-01, MA5-VOL-C-01

Content

Recognise the features associated with volume, capacity and mass

- Identify the language of volume, capacity and mass
- Identify features of an object associated with volume
- Identify features of an object associated with capacity
- Identify features of an object associated with mass

Estimate and measure volume, capacity and mass using informal and formal units

- Use informal units to estimate and measure volume, capacity and mass
- Recognise appropriate formal units for measuring volume, capacity and mass, including their abbreviations
- Estimate and measure volume, capacity and mass using formal units
- Compare volume and capacity using formal measurements
- Compare mass using formal units

Apply knowledge of estimating and measuring volume, capacity and mass to solve problems

- Compare the volume and capacity of different objects to solve problems
- Compare the mass of different objects to solve problems
- Apply volume, capacity and mass to solve problems

Area

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- measures and uses area in everyday contexts **MALS-ARE-01**

Related Stage 4/5 outcomes: MA4-ARE-C-01, MA5-ARE-C-01, MA5-ARE-P-01

Content

Recognise features of an object associated with area

- Identify language of area
- Recognise that area is the space taken up by a flat shape or surface

Estimate and measure area in informal and formal units

- Use informal units to estimate and calculate area
- Recognise appropriate formal units for measuring area, including their abbreviations
- Use formal units and tools for measuring area
- Calculate area in everyday contexts

Apply knowledge of estimating and measuring area to solve problems

- Estimate areas to solve problems
- Apply knowledge of area to solve problems

Geometry

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- explores 2-dimensional shapes and 3-dimensional objects **MALS-GEO-01**

Related Stage 4/5 outcomes: MA4-GEO-C-01, MA5-GEO-C-01

Content

Recognise the features of 2-dimensional shapes

- Identify properties of 2-dimensional shapes
- Recognise 2-dimensional shapes in the environment
- Sort 2-dimensional shapes based on attributes

Recognise features of 3-dimensional objects

- Identify properties of 3-dimensional objects
- Recognise 3-dimensional objects in the environment
- Sort 3-dimensional objects based on attributes
- Sort 3-dimensional objects based on attributes using formal language

Recognise similarities between 2-dimensional and 3-dimensional space across a range of contexts

- Identify common properties that 2-dimensional shapes share with 3-dimensional objects
- Use 2-dimensional shapes to construct a 3-dimensional object

Explore 2-dimensional space across a range of contexts

- Use 2-dimensional shapes to solve problems

Explore 3-dimensional space across a range of contexts

- Manipulate 3-dimensional objects to solve problems

Position

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- demonstrates knowledge of position and direction in everyday contexts **MALS-POS-01**

Related Stage 4/5 outcomes: MA4-LIN-C-01, MA5-LIN-C-01, MA5-LIN-C-02

Content

Recognise the language of position in everyday contexts

- Locate the position of familiar objects
- Respond to positional language to follow an instruction

Use the language of position

- Describe the position of 2 objects in relation to each other
- Describe the position of themselves in relation to their environment

Identify position on maps and plans

- Recognise that objects and features are represented on a map or a plan
- Locate familiar places on maps
- Recognise different cultural representations of maps

Recognise the language of direction

- Recognise the language of direction in everyday contexts
- Respond to directional language to follow an instruction

Use the language of direction

- Provide directions to get from one location to another

Use maps to demonstrate understanding of directions

- Use a map to follow directions between locations
- Use a map to approximate a location of a place in relation to another
- Locate current location on a map

Representing data

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- recognises and represents data in everyday contexts **MALS-DAT-01**

Related Stage 4/5 outcomes: MA4-DAT-C-01

Content

Recognise data in everyday contexts

- Recognise language used when speaking about data
- Recognise information from data displays
- Identify data displays from a variety of sources
- Demonstrate understanding of data representation

Collect data

- Select a method to collect data
- Participate in collecting data

Display data

- Sort collected data into groups
- Display collected data in a variety of formats
- Use appropriate strategies to construct data displays, with or without technology
- Apply conventions for displaying data

Interpreting data

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- interprets information from data displays **MALS-DAT-02**

Related Stage 4/5 outcomes: MA4-DAT-C-02, MA5-DAT-C-01, MA5-DAT-C-02

Content

Identify information from data displays

- Identify information presented in data displays
- Select data to answer questions

Draw conclusions from data displays

- Draw conclusions from information in data displays
- Select and organise the most relevant information to evaluate

Chance and probability

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- applies chance and probability to everyday events **MALS-PRO-01**

Related Stage 4/5 outcomes: MA4-PRO-C-01, MA5-PRO-C-01

Content

Recognise language of chance in everyday contexts

- Identify events that are certain and uncertain
- Use language of chance to indicate the likelihood of familiar events

Apply chance in everyday contexts

- Determine the likelihood of an event
- Predict the likelihood of a routine event
- Predict the likelihood of a non-routine event
- Order events according to their likelihood

Use chance and probability to reason in everyday contexts

- Apply the element of chance to make decisions about familiar events
- Recognise numerical values assigned to the probability of events occurring in everyday contexts
- Conduct experiments to determine the probability of an outcome
- Use knowledge of probability to make a prediction
- Apply knowledge of chance and probability to justify a prediction